Pre-Calculus

Tottaly Real Name

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Introduction

Message From The Author

Hello all. Welcome to your journey through pre-calculus, the foundation for a branch of mathematics that will prove essential to various professions/fields (physics, business, economics, engineering, etc.) Not only that, understanding the depths of calculus (and pre-calculus) will change your perspective on mathematics. I truly hope you'll appreciate the wealth pre-calculus has to offer and enjoy the process of making it your own.

A Little About Me

At the time of writing this text (March 2026), I'm a first-year community college student taking Multi-variable/Vector Calculus (Calc III). In the grand scheme of things, I'm not much further ahead in mathematics than you (if you're a student). But I thoroughly enjoy mathematics and want to show you the ins and outs of it. I hope I'm able to help you with your studies through my insights and explanations of concepts taught in pre-calculus.

Disclaimer

Before continuing with the material, I'd like to make a few things completely clear:

1. **This text is not a substitute for classroom instruction or your teacher/professor.** Your teachers and professors remain your **number one resource**. This text is intended only as a complimentary tool or additional material.

2. **I am not an expert and have no formal qualifications** (i.e., degrees or certificates) beyond a high school diploma. Although I will do my best to explain the concepts of pre-calculus, this is another important reason why you should not rely solely on this text.

Diagnostic

Let's do a diagnostic before you begin. This is supposed to show what you may need to review before continuing. Take your time with these questions. Aim for accuracy over speed. It's crucial that you understand how to solve these questions.

Problem 1

Given the definition of the function $h(x)$, evaluate $h(-2)$.

$$h(x) = 3x^3 - x^2 + 7$$

Problem 2

For the given quadratic equation:

1. Determine whether you can find the roots **without** the quadratic formula.
2. Find the roots of the quadratic.

$$y = x^2 + 3x - 15$$

Problem 3

Rewrite $\sqrt{63}$ in its simplest form.

Problem 4

Given the definition of $g(x)$, create a table of all integer x values from $x = -2$ to $x = 2$. Provide values in exact form (i.e. no decimals).

$$g(x) = \sqrt{x+7}$$

Problem 5

Daniel deposits \$1000 into a savings account at "Some Bank." Daniel's amount experiences an interest rate of 5% over the course of 2 years. How much money does Daniel's bank account have after 2 years if interest is compounded

1. Every year
2. Every month
3. Every day
4. Continuously (Hint: You need to use e).

Problem 6

Given the equation below, solve for x.

$$3x^3 + 14x = 12x^2$$

Hint: Cubic equations have 3 solutions. Do you remember complex/imaginary numbers? :)

Problem 7

Simplify the following as best you can.

$$\frac{x^2 - 4}{x - 2}$$

Solution. Solve for $h(-2)$ given that $h(x) = 3x^3 - x^2 + 7$.

$$\begin{aligned} h(-2) &= 3(-2)^3 - (-2)^2 + 7 && \text{Substitute } -2 \text{ for } x \\ &= 3(-8) - 4 + 7 && \text{Take the cube and square} \\ &= -24 - 4 + 7 && \text{Combine } -24 \text{ and } -4 \\ &= -28 + 7 && \text{Combine } -28 \text{ and } 7 \\ h(-2) &= -21 && \text{Solution} \end{aligned}$$

Solution. Given $y = x^2 + 3x - 15$, (a) determine whether you can solve **without** the quadratic formula and (b) solve for the roots.
Part A. A quadratic is solvable without the quadratic formula if the discriminant ($b^2 - 4ac$) is a perfect square. In this case, $a = 1$, $b = 3$, $c = -15$.

$$(3)^2 - 4(1)(-15) = 9 + 60 = 69$$

The discriminant is not a perfect square. Therefore, the quadratic formula must be used.
Part B. I'll now apply the quadratic formula.

$$\begin{aligned} x &= \frac{-3 \pm \sqrt{69}}{2(1)} && \text{Use Values from } \textbf{Part A} \\ &= \frac{-3 \pm \sqrt{69}}{2} \\ x &\approx -5.6535, 2.6535 && \text{Solution} \end{aligned}$$

Solution. Rewrite $\sqrt{63}$ in its simplest form.

1. Prime factorize 63: $63 = 9 \times 7 = 3^2 \times 7$.

2. Remove the squared term from the radicand: $\sqrt{3^2 \times 7} = \sqrt{3^2} \times \sqrt{7} = 3\sqrt{7}$.

$$\boxed{\sqrt{63} = 3\sqrt{7}}$$

Solution. Create a table of integers from $x = -2$ to $x = 2$ for $g(x) = \sqrt{x+7}$. All values must be in exact form.

x	$g(x)$
-2	$\sqrt{5}$
-1	$\sqrt{6}$
0	$\sqrt{7}$
1	$\sqrt{8} = 2\sqrt{2}$
2	$\sqrt{9} = 3$

Solution. Daniel's $1000 experiences 5% interest over the course of 2 years.

I'll use the equation for exponential growth to represent Danial's bank account.

$$A = P\left(1 + \frac{r}{n}\right)^{tn}$$

where,

- A is the final amount
- P is the original amount deposited
- r is the interest rate
- n is the compounding rate
- t is years

If Compounded Yearly.

$$\begin{aligned}
A &= 1000(1+0.05)^2 && \text{Solve } 1+0.05 \\
&= 1000(1.05)^2 && \text{Evaluate } 1.05^2 \\
&= 1000(1.1025) && \text{Evaluate } 1000(1.1025) \\
A &= 1102.5 && \text{Solution}
\end{aligned}$$

If Compounded Monthly.

$$\begin{aligned}
&A = 1000\left(1 + \frac{0.05}{12}\right)^{2(12)} && \text{Simplify exponent} \\
&A = 1000\left(1 + \frac{0.05}{12}\right)^{24} && \text{Rewrite 1 as } \frac{12}{12} \\
&A = 1000\left(\frac{12}{12} + \frac{0.05}{12}\right)^{24} && \text{Evaluate } \frac{12+0.05}{12} \\
&A = 1000\left(\frac{12.05}{12}\right)^{24} && \text{Approximate with calculator} \\
&A \approx 1104.94 && \text{Solution}
\end{aligned}$$

If Compounded Daily.

$$
\begin{aligned}
A &= 1000\left(1+\frac{0.05}{365}\right)^{2(365)} && \text{Simplify exponent}\\
A &= 1000\left(1+\frac{0.05}{365}\right)^{730} && \text{Rewrite 1 as } \frac{365}{365}\\
A &= 1000\left(\frac{365}{365}+\frac{0.05}{365}\right)^{730} && \text{Evaluate } \frac{365+0.05}{365}\\
A &= 1000\left(\frac{365.05}{365}\right)^{730} && \text{Approximate with calculator}\\
A &\approx 1105.16 && \text{Solution}
\end{aligned}
$$

If Compounded Continuously. Continuously compounding interest rate is defined by the equation $A = Pe^{rt}$.

$$
\begin{aligned}
A &= 1000e^{0.05(2)} && \text{Simplify Exponent}\\
A &= 1000e^{0.1} && \text{Use Calculator For Approximation}\\
A &\approx 1105.17 && \text{Solution}
\end{aligned}
$$

Solution. Solve $3x^3 + 14x = 12x^2$ for x.

$$\begin{aligned} 3x^3 - 12x^2 + 14x &= 0 && \text{Moved } 12x^2 \text{ to the left} \\ x(3x^2 - 12x + 14) &= 0 && \text{Factored out an } x \end{aligned}$$

One solution is immediately clear because of the **Zero Product Property**: $x = 0$. The quadratic factor is not factorable, so we need to use the quadratic equation.

$$\begin{aligned} x &= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(3)(14)}}{2(3)} && \text{Plugging everything in} \\ &= \frac{12 \pm \sqrt{144 - 168}}{6} && \text{Multiplying everything out} \\ &= \frac{12 \pm \sqrt{-24}}{6} && \text{Simplifying radicand} \\ &= \frac{12 \pm 2i\sqrt{6}}{6} && \text{Expanding and simplifying radicand} \\ x &= 2 \pm \frac{\sqrt{6}}{3}i && \text{Dividing numerator by denominator} \end{aligned}$$

Thus, the solutions are

$$\boxed{x = 0, 2 + \frac{\sqrt{6}}{3}i, 2 - \frac{\sqrt{6}}{3}i}.$$

Solution. Simplify $\dfrac{x^2-4}{x-2}$.

$$\frac{(x-2)(x+2)}{x-2} \qquad \text{Use difference of squares}$$

$$x+2 \qquad \text{Divide numerator by } x-2;\ x \neq 2.$$

The final answer is

$$\boxed{\frac{x^2-4}{x-2} = x+2}.$$

If, after completing the diagnostic, you get 3 or more questions wrong, I highly encourage you to review some material from Algebra II. Don't feel down if you did, though. This is probably stuff you haven't seen for a few weeks or monthly (maybe longer). Maybe you're a bit rusty right now or it's just not your day. Just know that you'll need to understand and have a decent grasp of this before you're able to dive into pre-calculus.

Part I

Functions

Welcome. If you're here now, it's probably because you got 4 or more questions right on the diagnostic. Well done. That means you should have an understanding of how to manipulate equations, factor quadratics, solve word problems, work with exponential equations, simplify, etc. If that's not you (i.e. you get less than 4 questions right), you can still continue reading, but this *may* be a bit more challenging for you. Regardless of what boat you're in, I'll do my best to explain concepts as simply as possible.

Functions are by far the most important concept in pre-calclus and calculus. You'll continue to use functions *far* into calculus and even beyond calculus.

Chapter Overview

In no particular order, this chapter on functions aims to:

- Solidify your understanding of Domain and Range
- Help you feel comfortable with function transformations.
- Teach you how to work with composite functions.
- Solidify your understanding of inverses.
- Teach you piecewise functions.
- Familiarize you with rational functions and asymptotes
- Solidify your understanding of exponential and logarithmic functions.

Chapter 1

Domain and Range

You may have heard about domain and range while in Algebra II (or Integrated III) in the context of inverse functions. For example, you'll be given a function $f(x)$, ask to find it's inverse, $f^{-1}(x)$, and then to state the domain and range of $f(x)$ and $f^{-1}(x)$. Don't get me wrong, understand that is still essential. But the goal here is to really dive into what exactly the domain and range of a function are.

Definition

For any function, $f(x)$, the **domain** of that function are all values that, when passed into $f(x)$, result in a real output for which $f(x)$ is defined (i.e. excludes things like division by 0). This definition specifically applies to graphs on the xy-plane.

A very good example of this is the function

$$f(x) = \sqrt{x}.$$

This function has a domain of $0 \leq x < \infty$ or $x \in [0, \infty)$ (interval notation). This is the case because plugging any values less than 0 would result in a negative number under the radical, which we know is an imaginary number.

$$f(-1) = \sqrt{-1} = i \implies -1 \text{ is } \textbf{not} \text{ in the domain.}$$

Definition

For any function, $f(x)$, the **range** of that function are all possible outputs given the **domain restriction**.

Looking back at our example, we can see that the range of the function, $f(x)$ is restricted to positive numbers. Plugging a negative number results in a complex solution, which **cannot be plotted on a standard coordinate plane consisting of x and y axes**. Therefore, we can say that the range of the function, $f(x)$, is $0 \leq y < \infty$ or $y \in [0, \infty)$ (interval notation).

Another example we can use is

$$g(x) = \frac{1}{x}.$$

Recall that a 0 in the denominator makes the expression **undefined**. For that reason, the domain is $x \neq 0$ or $x \in (-\infty, 0) \cup (0, \infty)$. For now, I'll simply give that the range of this function is $y \neq 0$ or $y \in (-\infty, 0) \cup (0, \infty)$. To see why, we need to understand a concept called an **asymptote**.

Before that, let's do a practice problem.

Problem

Determine the domain and range of the function $f(x) = \sqrt{2x+1}$.

We haven't covered how to find domains. But, if you think you can figure it out, give it a shot. Regardless, here's the solution.

Solution. We know that the value inside a radical can never be below 0. So, to find our domain restriction, we need to solve what's under the radical for 0.

$$\begin{aligned} 2x + 1 &= 0 \\ 2x &= -1 \\ x &= -\frac{1}{2} \end{aligned}$$

The work we just did shows that any input less than $-\dfrac{1}{2}$ will result in a negative under the radical. That means $-\frac{1}{2}$ the minimum value that we can plug into this function. Thus, our domain is $x \geq -\frac{1}{2}$.

Now that we have our domain, we can find our range by seeing what value we get when we plug values on the edges of our domain restrictions. However, in this case, we already know what our range is. Substituting $x = -\frac{1}{2}$ gives the lowest valid number under the radical: 0. So out range is $y \geq 0$.

1.1 Asymptotes

To understand what exactly an asymptote is, let's try plugging some values into the function $g(x)$ and see what happens. First, let's try $x = 1$.

$$g(1) = \frac{1}{1} = 1$$

Now let's try plugging 10.

$$g(10) = \frac{1}{10} = 0.1$$

Let's continue adding zeroes and plugging into the function.

$$\begin{aligned} g(100) &= \frac{1}{100} = 0.01 \\ g(1000) &= \frac{1}{1000} = 0.001 \\ g(10000) &= \frac{1}{10000} = 0.0001 \\ g(1000000000) &= \frac{1}{1000000000} = 0.000000001 \end{aligned}$$

As we increase the number we input into the function, the smaller the result gets. The same is true with negative numbers:

$$\begin{aligned} g(-100) &= -\frac{1}{100} = -0.01 \\ g(-1000) &= -\frac{1}{1000} = -0.001 \\ g(-10000) &= -\frac{1}{10000} = -0.0001 \\ g(-1000000000) &= -\frac{1}{1000000000} = -0.000000001 \end{aligned}$$

You may have noticed that, no matter how large or small of a number we input into $g(x)$, the answer will never reach 0. That's because the numerator is nonzero—it's the number 1 in this case. The answer to 1 divided by anything will **never** be 0. We can express this idea in a slightly different way. We can say that the function of $g(x)$ will never touch the line $y = 0$. That is, the function will never have a point with a y-value of 0. As a result of this new definition, we can say that the function $g(x)$ has an asymptote line of $y = 0$.

Think about what happens as you decrease the value of x inputted. Remember that dividing by a number less than 1 increases the value. For example, if we plug $x = 0.1$ into the function, we get

$$g(0.1) = \frac{1}{0.1} = 10.$$

If we continue decreasing the number, we see that the value of $g(x)$ grows without bound.

$$\begin{aligned} g(0.01) &= \frac{1}{0.01} = 100 \\ g(0.001) &= \frac{1}{0.001} = 1000 \\ g(0.0001) &= \frac{1}{0.0001} = 10000 \\ g(0.000000001) &= \frac{1}{0.000000001} = 1000000000 \end{aligned}$$

The same also applies with negative numbers.

$$\begin{aligned} g(-0.01) &= -\frac{1}{0.01} = -100 \\ g(-0.001) &= -\frac{1}{0.001} = -1000 \\ g(-0.0001) &= -\frac{1}{0.0001} = -10000 \\ g(-0.000000001) &= -\frac{1}{0.000000001} = -1000000000 \end{aligned}$$

This tells us that, as $g(x)$ gets closer and closer to 0, the value given by $g(x)$ goes to ∞ if the inputs are positive and $-\infty$ if the inputs are negative. Because the value of x can never be 0, we can say that the function $g(x)$ has another asymptote line of $x = 0$.

Definition

An **asymptote** is a line that a graph approaches as x either grows very large ($x \to \infty$ or $x \to -\infty$) or approaches a particular value. More formally:

- The line $x = a$ is a **vertical asymptote** of f if $\lim_{x \to a} f(x) = \pm\infty$.
- The line $y = L$ is a **horizontal asymptote** of f if $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$.

The graph gets extremely close to the asymptote. The asymptote really only cares for **extremely** large values, so the function actually *can* cross a **horizontal** asymptote before it starts to slowly approach it. This is **not true** for a vertical asymptote.

There are a few kinds of asymptotes. The three I'll cover here are the **horizontal**, **vertical**, and **slant** asymptotes.

1.1.1 Vertical Asymptotes

Vertical asymptotes come about because of domain restrictions. Not all domain restrictions result in vertical asymptotes, but **all vertical asymptotes are a result of a domain restriction**. For example, the function $g(x) = \frac{1}{x}$ has a vertical asymptote at $x = 0$ because the function $g(x)$ has $x = 0$ excluded from its domain.

Because of this fact, we can find vertical asymptotes by solving for the domain restriction of a function and observing/determining its behavior as the function approaches the restriction. That is the process we used when first defining an asymptote in the previous section (plugging numbers closer and closer to 0 and seeing what happens).

Let's try this on a practice problem.

Problem

For the function, f, defined below, find it's vertical asymptotes.

$$f(x) = \frac{1}{(x-1)(x+3)}$$

This problem can be a little tricky, but you can solve it by remembering how to find the domain restriction of a function (dicussed in the earlier section). Read ahead to the solution when you're done or get stuck.

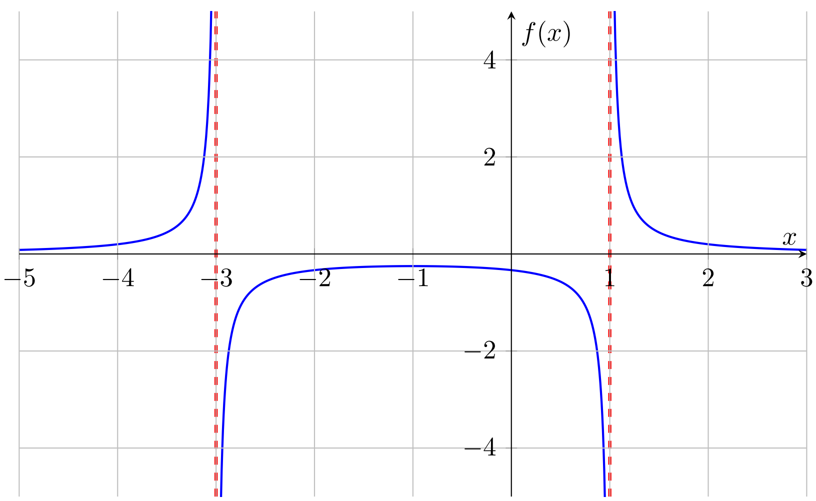
Solution. To find the domain restriction of a function, we need to see if there's anything that could potentially "break" the function. In this case, we see that we have a fraction. That means we can break the function by making the denominator equal to 0. To see what values of x make the denominator equal to 0, we can set it equal to 0 and solve for x.

$$(x-1)(x+3) = 0$$

Using the **Zero Product Property** (ZPP), we can see that this quadratic evaluates to $x = -3, 1$. Thus, $x = -3, 1$ are restricted from our domain. Now, we can see what happens to the function when we plug numbers extremely close to our domain restrictions.

$$\begin{aligned} f(-3.0000001) &\approx 2499999.938 \\ f(-2.9999999) &\approx -2500000.063 \\ f(0.99999999) &\approx -25000000.06 \\ f(1.00000001) &\approx 24999999.94 \end{aligned}$$

We can see that plugging numbers close to our domain restrictions result in enormous numbers. This means that our asymptotes must be the lines $x = -3$ and $x = 1$.



Problem

For the function, g, define below, find it's vertical asymptotes.

$$g(x) = \frac{3}{x^2 + 1}$$

Give it a shot. When you're done or get stuck, read ahead to the solution.

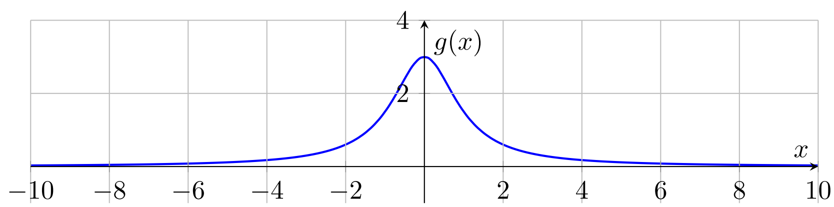
Solution. Just like before, we can see that the denominator cannot equal 0. So we can set it to 0 and solve.

$$x^2 + 1 = 0$$

However, there's something different about this solution. If we let $a = 1$, $b = 0$, and $c = 1$, we can see that the discriminant $(b^2 - 4ac)$ is less than 0.

$$0 - 4(1)(1) = -4$$

This means that there are no real solutions to this quadratic. When the denominator set equal to 0 has no real solutions, it means the function has no vertical asymptotes.



1.1.2 Horizontal Asymptotes

Horizontal asymptotes come about because of a range restriction. To be more accurate, horizontal asymptotes are a result of the end behavior of a function as x approaches $\pm\infty$. Just like with vertical asymptotes, we can solve for the range of a function and observe its behavior as it aproaches the range restrictions. Conversely, we can use values from the domain that, when plugged into the function, gradually bring the function closer and closer to its range restriction.

What does this mean? Take the example $g(x) = \frac{1}{x}$. Earlier in the section on asymptotes ("1.1 Asymptotes"), we calculated $g(x)$ for extremely large values of x. As the value of x increased, the output $g(x)$ decrased. In fact, it slowly got closer and closer to its domain restriction: $y \neq 0$. Let's look at some example problems.

Problem

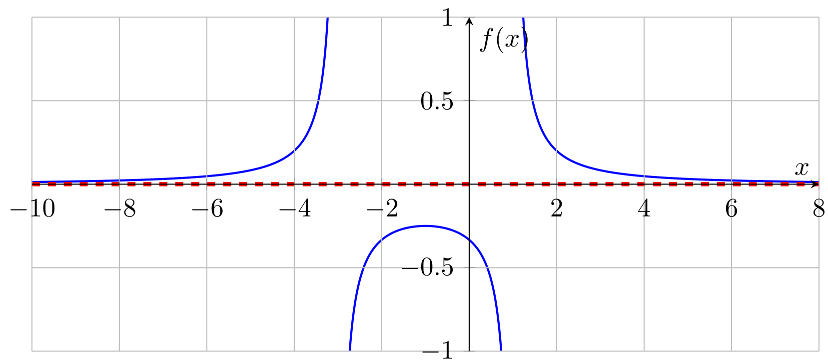
For the function, f, defined below, find it's horizontal asymptotes.

$$f(x) = \frac{1}{(x-1)(x+3)}$$

Solution. The function $f(x)$ cannot equal 0, regardless of what the denominator is, because the numerator will **always** be 1. Therefore, we can say that this method has a range of $y \neq 0$. To see if this is indeed a horizontal asymptote, let's plug some large values and see what happens.

$$\begin{aligned} f(100) &\approx 0.0000981 f(10000) &&\approx 0.000000009998 \\ f(-100) &\approx 0.000102 f(-10000) &&\approx 0.00000001 \end{aligned}$$

We can see that plugging large values of x gets us closer and closer to our range restriction $y \neq 0$. Since that is our only range restriction, we can say that the function $f(x)$ has a horizontal asymptote of $y = 0$.



Problem

For the function, g, define below, find it's horizontal asymptotes.

$$g(x) = \frac{3x^2 + 5}{x^2 + 1}$$

Your first instinct would be to look for the range. While that *is* a valid strategy, you'll notice that it's quite cumbersome. I mean... Where do you even start? Throwing a bunch of large values of x *would* give the answer, but let's think of a better way to explain this. It's crucial that we explore this further because it reveals a trick we can use to solve any problem like this in an instant.

Solution. First, let's plug the number 1 into this function.

$$g(1) = \frac{3(1)^2 + 5}{(1)^2 + 1} = \frac{3 + 5}{1 + 1}$$

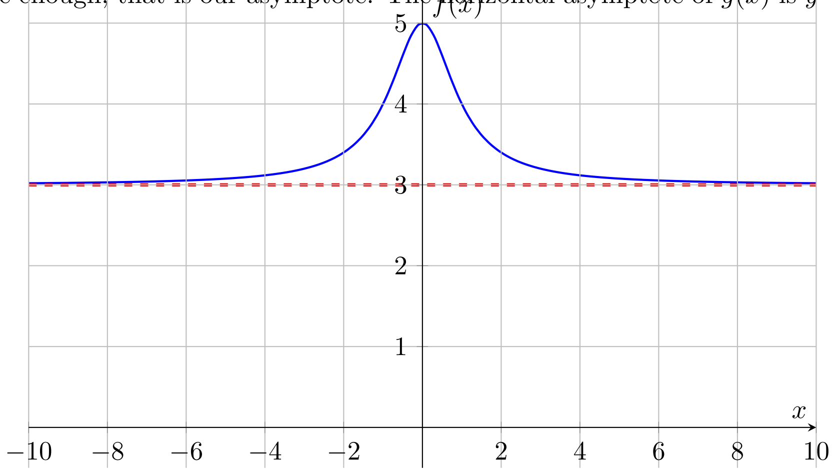
When we plug $x = 1$ into this function, that $+5$ and $+1$ in the numerator and denominator *really* change the answer. We're jumping from 5 to 8 in the numerator and from 1 to 2 in the denominator. Now, let's try plugging a larger number, like 100.

$$g(100) = \frac{3(100)^2 + 5}{(100)^2 + 1} = \frac{3(10000) + 5}{10000 + 1} = \frac{30000 + 5}{10000 + 1}$$

Take a look at this. The +5 and +1 in the numerator and denominator don't really do anything anymore. Yeah, they increase the number, but compared to how large 30000 and 10000 are, they're basically nothing. It should be clear, now, that plugging larger and larger numbers into x makes the +5 and +1 more and more insignificant. As x increases, we're basically looking at

$$g(x) = \frac{3x^2}{x^2} = \boxed{3}$$

Sure enough, that is our asymptote. The horizontal asymptote of $g(x)$ is $y = 3$.



If the degree of the numerator and denominator are the same, the horizontal asymptote is determined by dividing the coefficient of the highest degree in the numerator by that of the denomianator.

1.1.3 Polynomial Asymptotes

Polynomial asymptotes, unlike vertical and horizontal, are quite tricky. They only occur when the degree of the polynomial in the numerator is greater than that in the denominator. It's not easy to define them using a domain and range because they don't conform strictly to the domain (vertical asymptote) or range (horizontal asymptote). We must, instead, be a little more creative.

Let's remember what division does to a number. Take something like this, for example: $10 \div 3$. In many cases, the number you're dividing by, the divisor, doesn't fit evenly into the number you're dividing, the dividend. In such cases, you are left with a remainder that can be written as a fraction, like so:

$$3 + \frac{1}{3}.$$

The answer we get is such that multiplying it by 3 gets us back to the original number, 10. This is true with polynomials, too.

If we allow the following to be true:
$f(x)$ - Dividend
$g(x)$ - Divisor
$q(x)$ - Quotient
$r(x)$ - Remainder

Then we can say that the solution to polynomial long division can be expressed like so:

$$\frac{f(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}$$

If we let:
$f(x) = x^2$
$g(x) = x + 1$

We get the following solution:

$$
\begin{array}{r|lll}
 & x & -1 & \\ \cline{2-4}
x+1 & x^2 & 0 & \\
 & -(x^2 & x) & \\ \cline{2-3}
 & & -x & 0 \\
 & & -(-x & -1) \\ \cline{3-4}
 & & & 1
\end{array}
$$

In this case, $q(x) = x - 1$ and $r(x) = 1$. Thus, we can write the solution

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}.$$

Notice how, in the $\frac{r(x)}{g(x)}$ term, the x is in the denominator. As x grows larger and larger, like to 10000 or ∞, that term becomes smaller and smaller.

$$\frac{\text{small number}}{\text{big number}} = \text{small number}$$

As that term reaches 0, we are left with $q(x)$, which we established will never actually reach the rational function defined as $\frac{f(x)}{g(x)}$. So, no matter what values of x we put in there, and no matter how close $q(x)$ might get to $\frac{f(x)}{g(x)}$, it will never touch it. **That is the definition of an asymptote.**

Definition

Given a dividend, $f(x)$, and a divisor of a lower degree, $g(x)$, written as a rational function, $y = \frac{f(x)}{g(x)}$, the polynomial asymptote is defined as

$$y = q(x),$$

where $q(x)$ is the quotient obtained from dividing $\frac{f(x)}{g(x)}$.

While this is true for all rational functions with polynomials in the numerator and denominator, the one you'll most commonly see is a rational function where the degree of the numerator is 1 greater than the denominator. This is also called a **slant** or **oblique** asymptote.

Let's look at a practice problem together.

Problem

Find all the asymptotes for the rational function

$$f(x) = \frac{3x^2 + x - 10}{x - 5}.$$

*Note that $f(x)$, in this case, is the entire rational function and **not** the dividend.*

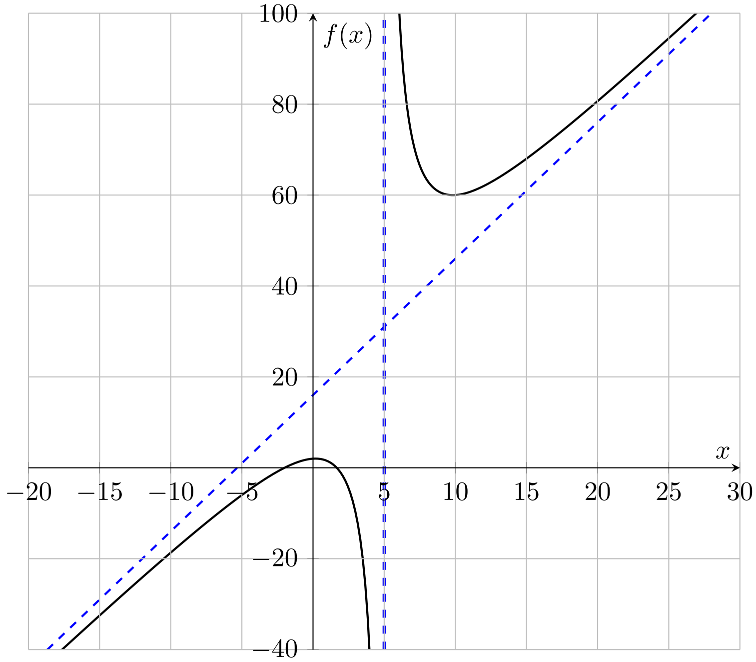
Solution. We need to find the quotient of this rational function. Beacuse the denominator has a degree of 1, we can use synthetic division.

$$\begin{array}{r|rrr} 5 & 3 & 1 & -10 \\ & & 15 & 90 \\ \hline & 3 & 16 & 80 \end{array}$$

$$q(x) = 3x + 16$$

That's the first asymptote. There is a second hiding in the denominator. If we set that equal to 0, we can solve for a vertical asymptote as well.

$$x - 5 = 0 \Longrightarrow x = 5$$


100
80
60
40
20
−20
−40
$f(x)$
x
−20 −15 −10 −5 5 10 15 20 25 30

Chapter 2

Transformations

Up to this point, we've just been drawing functions and finding key details about them (asymptotes, domain, range). Now it's time to move functions around and warp them in all kinds of ways. You probably learned how to do this back in Algebra II (or Integrated III). The goal here is to really cement that understand.

If $f(x)$ is the parent function, these are what the transformations look like:

$$af\left(\frac{1}{b}(x-h)\right)+k$$

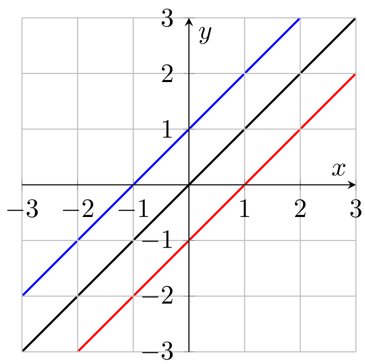
- h - Horizontal shift (positive = right)
- k - Vertical Shift (positive = up)
- a - Stretching and Shrinking along the vertical axis
  - Larger Values = More Vertical Stretching
- b - Stretching and Shrinking along the horizontal axis
  - Larger Values = More Horizontal Stretching

Note: Some texts use b instead of $\frac{1}{b}$. When using b, larger values of b horizontally ***compress*** *instead of stretch.*

Let's try to understand why each of these variables transform the function the way they do.

2.1 Vertical Shifts

The easiest of the transformations to consider is k. This variable vertically shifts the function by its value. For example, if $f(x) = x$, then $g(x) = x + 1$, where $k = 1$, is simply the function $f(x)$ but moved up a unit. In the same manner, a function $h(x) = x - 1$, where $k = -1$, is the function $f(x)$ moved down a unit.



$f(x) = x$ (Black), $g(x) = x + 1$ (Blue), $h(x) = x - 1$ (Red)

A vertical shift moves all parts of a funtion the specified number of units up or down. That includes asymptotes. For example, if the function I'll redefine as $f(x) = \dfrac{1}{x}$ has a horizontal asymptote of $y = 0$, the new redefined function $g(x) = \dfrac{1}{x} + 1$ wil have a horizontal asymptote of $y = 1$.

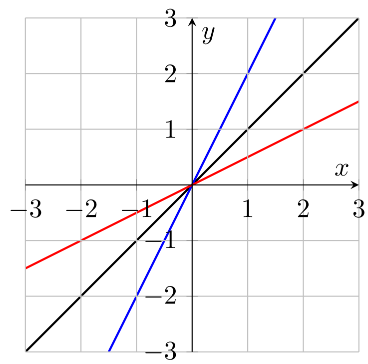
Definition

Given a function $f(x)$ and a vertical shift k, the new function $g(x) = f(x) + k$ is the function $f(x)$ vertically translated k units.

- When $k < 0$, $f(x)$ moves in the negative y direction (down).
- When $k > 0$, $f(x)$ moves in the positive y direction (up).
- When $k = 0$, $f(x) = g(x)$. The function does not move.

2.2 Vertical Stretches

A close second in terms of simplicity is the vertical stretch, represented by the letter a. This function multiplies every output of a function by its value. For example, if $f(x) = x$, the function $g(x) = 2x$, where $a = 2$, means that every value plugged into $g(x)$ is doubled. Likewise, if we set $a = \frac{1}{2}$, every value plugged into a new function $h(x) = \frac{1}{2}x$ is halved.



$f(x) = x$ (Black), $g(x) = 2x$ (Blue), $h(x) = \frac{1}{2}x$ (Red)

Note that a for a linear function is its slope.

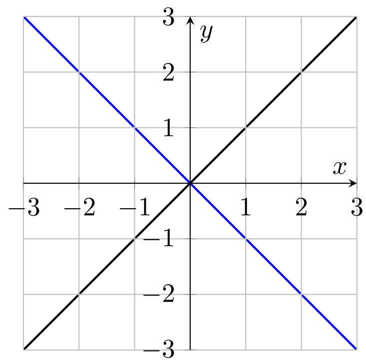
Definition

Given a function $f(x)$ and a vertical stretch a, the new function $g(x) = af(x)$ is the function where every value in the range of f is multiplied by a.

- When $a > 1$, stretching occurs.
- When $0 < a < 1$, compression occurs.
- When $a = 1$, the function is unaffected.
- When $a = 0$, the function is a horizontal line.

2.2.1 Vertical Reflection

A vertical reflection is achieved when a is negative. A negative value for a means every value for a function $f(x)$ that used to be positive becomes negative. For example, when $a = 1$ for the function $f(x) = x$, inputting $x = 1$ produces $y = 1$. However, for a function $g(x) = -x$, where $a = -1$, plugging $x = 1$ produces $y = -1$.



$f(x) = x$ (Black), $g(x) = -x$ (Blue),

Definition

Given a function $f(x)$, a vertical reflection, also called a reflection along the $x-axis$ or $y = 0$, occurs when $a < 0$. For the function $g(x) = af(x)$, all values in the range are switched from positive to negative (and vice versa), and then stretched by $|a|$.

- When $a < -1$, vertical stretching in the negative direction occurs.
- When $-1 < a < 0$, vertical compressing in the negative direciton occurs.
- When $a = -1$, the function is reflected without stretching or compressing.

2.3 Horizontal Shifts

Now for the slightly trickier transformations. This first one is the horizontal shift, represented by the letter h. A horizontal shift, like a vertical one, moves the function left or right. For example, if $f(x) = x^2$, a horizontal shift one unit to the right would result in a new function $g(x) = (x-1)^2$, where $h = 1$.

Herein lies the confusion: If we're subtracting x by 1, why does the function move in the positive x direction? Aren't negative values of x on the left?

Negative values *are* on the left, but it shifts to the right because the result that would've normally been produced by $f(x)$ is now produced by some larger input of x.

$f(2) = 4$ because when you square 2 you get 4. But if you want $g(x)$ to also produce 4, you have to use $g(3)$.

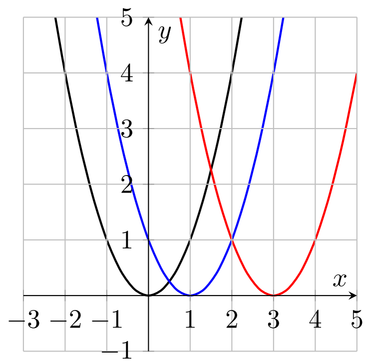
$$g(3) = (3-1)^2 = (2)^2 = 4$$

To produce the same result, you had to input a larger value of x. The best example of this is getting the functions to result in 0.

$$\begin{aligned} &f(0) = (0)^2 = 0 && \text{Point at } (0,0) \\ &g(1) = (1-1)^2 = (0)^2 = 0 && \text{Point at } (1,0) \end{aligned}$$

Notice that the difference for when $f(x) = g(x) = 4$ and $f(x) = g(x) = 0$, the difference in input for $f(x)$ and $g(x)$ is the value of h. It can then be proposed that increasing the value of h to, say, $h = 3$, plugging 3 into $h(x) = (x-3)^2$ will produce 0.

$$h(3) = (3-3)^2 = (0)^2 = 0 \qquad \text{Point at } (3,0)$$



$f(x) = x^2$ (Black), $g(x) = (x-1)^2$ (Blue), $h(x) = (x-3)^2$ (Red)

Definition

Given a function $f(x)$ and a horizontal shift k, the new function $g(x) = f(x-1)$ is the function of $f(x)$ horizontally translated h units.

- When $h < 0$, $f(x)$ moves in the negative x direction (left).
- When $h > 0$, $f(x)$ moves in the positive x direction (right).
- When $h = 0$, $f(x) = g(x)$. The function does not move.

2.4 Horizontal Stretching

Horizontal stretching is by far the most confusing. At the beginning of this chapter, I wrote $\frac{1}{b}(x-h)$ and noted that some texts use b instead of $\frac{1}{b}$. In this section, I'll be sure to clarify exactly what adding a value in front of those parentheses does to the function and explain the differences between b and $\frac{1}{b}$.

Let's define a function $f(x) = x^2$. I'll now create a table with three columns: one for the x value plugged into the function, one for the value of x after b is applied, and one for the output of the function. In this case, $b = 1$, so the table of $f(x)$ looks something like this:

x	$1x$	$f(x)$
-2	-2	4
-1	-1	1
0	0	0
1	1	1
2	2	4

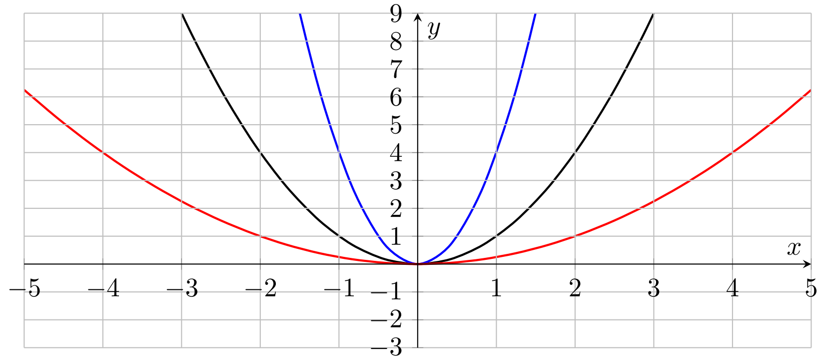
Now let's look at another function $g(x) = (2x)^2$.

x	$2x$	$g(x)$
-2	-4	16
-1	-2	4
0	0	0
1	2	4
2	4	16

Notice how an input of 2 all of a sudden becomes 4. This is *almost* like a vertical stretch. If this *had* been a vertical stretch (i.e. $y = 2x^2$), an input of 2 would produce $y = 8$. But we got 16 this time with $g(x)$. Instead of imagining it as a stretch upward, imagine the graph being smushed together so that the points are more tightly packed near the y-axis line ($x = 0$).

Btw, you could totally see it as something similar to a vertical stretch. Your teacher might not like it (and neither do I) because doing so would make it easy to confuse the vertical stretch $2x^2$ with the "vertical stretch" $(2x)^2$. So, just for teacher's and your own sanity, just think of $(2x)^2$ as a horizontal compression.

Let's look at these functions on a graph!



$$f(x) = x^2 \text{ (Black)},\ g(x) = (2x)^2 \text{ (Blue)},\ h(x) = \left(\frac{1}{2}x\right)^2 \text{ (Red)}$$

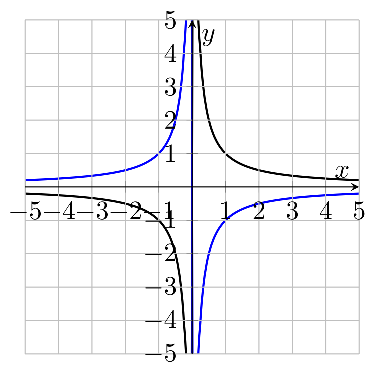
Definition

Given a function $f(x)$ and a horizontal stretch b, the new function $g(x) = f\left(\frac{1}{b}x\right)$ is the function of f horizontally stretched by a factor of b.

- When $b > 1$, horizontal stretching occurs.
- When $0 < b < 1$, horizontal compression occurs.
- When $b = 1$, the function is unaffected.
- When $b = 0$, the function is undefined.

2.4.1 Horizontal Reflections

A horizontal reflection is achieved when b is negative. A negative value for b means every input for a function $f(x)$ that used to be positive becomes negative. For example, when $b = 1$ for the function $f(x) = \frac{1}{x}$, inputting $x = 1$ produces $y = 1$. However, for a function $g(x) = \frac{1}{-x}$, where $b = -1$, plugging $x = 1$ produces $y = -1$.



$f(x) = \frac{1}{x}$ (Black), $f(x) = \frac{1}{-x}$ (Blue),

Definition

Given a function $f(x)$, a vertical reflection, also called a reflection along the y-axis or $x = 0$, occurs when $b < 0$. For the function $g(x) = f\left(\frac{1}{b}x\right)$, all values in the domain are switched from positive to negative (and vice versa), and stretched by a factor of $|b|$.

- When $b < -1$, horizontal stretching in the negative direction occurs.
- When $-1 < a < 0$, horizontal compressing in the negative direction occurs.
- When $a = -1$, the function is reflected without stretching or compressing.

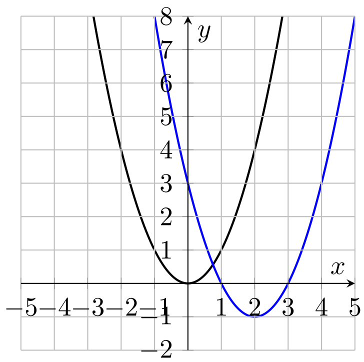
2.5 Practice Problems

Let's do some practice problems, shall we? I'll only give a couple, each increasing in difficulty.

Problem

Let the function f be $f(x) = x^2$. Translate (move) this function 2 units to the right and 1 unit down. Write your answer as a new function g.

Solution. We're given two translations: A horizontal translation h and a vertical translation k. We're asked to move this function two units to the right, meaning $h = 2$. We're then asked to move this function 1 unit down, meaning $k = -1$. Apply these to the template $g(x) = f(x - h) + k$. Plugging these into the function yields the answer $\boxed{g(x) = (x - 2)^2 - 1}$.



$f(x) = x^2$ (Black), $g(x) = (x - 2)^2 - 1$ (Blue),

Problem

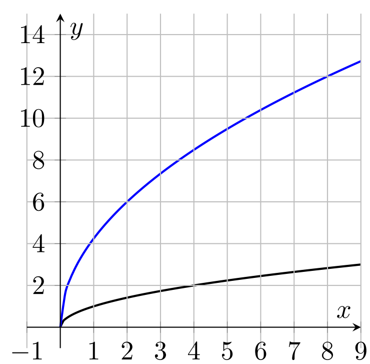
Let the function f be $f(x) = \sqrt{x}$. Apply a vertial stretch with a factor of 3 and horizontal stretch with a factor of $\frac{1}{2}$. Write your answer as a new function g.

Solution. We're given two transformations: A vertial stretch a and a horizontal stretch b. We're asked to vertically stretch this function by a factor of 3, meaning $a = 3$, and horizontally stretch this function by a factor of $\frac{1}{2}$, meaning $b = \frac{1}{2}$. Apply these to the template $g(x) = af\left(\frac{1}{b}x\right)$.

The vertical stretch is easy, just slap the 3 in for a. The horizontal stretch, on the other hand, is more tricky. We know that $b = \frac{1}{2}$, but plugging that in gives $\frac{1}{1/2}$. This is a complex fraction and serves only to complicate things. To get rid of this, we multiply the numerator and denominator by the reciporical of the denominator.

$$\frac{1}{\frac{1}{2}} \cdot \frac{\frac{2}{1}}{\frac{2}{1}} = 1 \cdot \frac{2}{1} = 1 \cdot 2 = 2$$

Thus, our final answer is $\boxed{g(x) = 3\sqrt{2x}}$.



$f(x) = x^2$ (Black), $g(x) = (x-2)^2 - 1$ (Blue),

Problem

Let the function f be $f(x) = \dfrac{1}{x}$. Translate the function 3 units to the left and 5 units up. Transform the function with a vectical stretch factor of 2 and horizontal stretch factor of 3. Reflect the function over the y-axis. Write your answer as a new function g.

Solution. There are a lot of things we need to do to this function, so let's list them one at a time.

- Move 3 units left ($h = 3$).
- Move 5 units up ($k = 5$).
- Vertically stretch by a factor of 2 ($a = 2$).
- Horizontally stretch by a factor of 3 ($b = 3$).
- Reflect over the y-axis (We'll worry about this later).

With what we've gathered so far, we can create a template:

$$g(x) = 2f\left(\frac{1}{3}(x-3)\right) + 5$$

From here, we can plug everything in

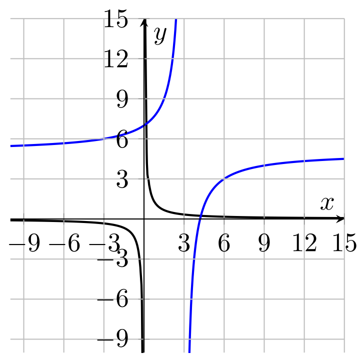
$$g(x) = 2\left(\frac{1}{\frac{1}{3}(x-3)}\right) + 5$$

Now we can reflect over the y-axis. Remember that the y-axis is the line $x = 0$. It's the vertical line that goes straight through the xy-plane. Since we're reflecting from left to right, we must multiply b by -1. This gives

$$g(x) = 2\left(\frac{1}{-\frac{1}{3}(x-3)}\right) + 5.$$

Now, we begin solving:

$$\begin{aligned} g(x) &= \frac{2}{-\frac{1}{3}(x-3)}\left(\frac{\frac{3}{1}}{\frac{3}{1}}\right)+5 \\ &= \frac{6}{-(x-3)}+5 \\ &= 5-\frac{6}{x-3} \end{aligned}$$



$f(x) = \frac{1}{x}$ (Black), $g(x) = 5 - \frac{6}{x-3}$ (Blue),

Note that you could also write this function as

$$g(x) = \frac{6}{3-x} + 5.$$

and get the same result. That's how I'd write it. You could also just leave it as

$$g(x) = \frac{6}{-(x-3)} + 5 \Longrightarrow \boxed{g(x) = -\frac{6}{x-3} + 5}.$$

It's all stylistic/presentation preferences.

Chapter 3

Function Composition

Function composition is esentially the operations (i.e. addition, subtraction, multiplication, division) with functions instead of numbers and variable. It may *seem* intimidating, but it's really super easy. I'll be sure to take it one step at a time. But you'll realize that it's really just the basic operations written differently. In this section, I'll be covering:

- Function addition $(f+g)(x)$
- Function subtraction $(f-g)(x)$
- Function multiplication $(f \cdot g)(x)$
- Function division $\left(\frac{f}{g}\right)(x)$

Let's begin with addition, since it's the simplest one.

3.1 Function Addition

Function addition is exactly what it sounds like. Given a function $f(x)$ and $g(x)$, you just add them together. So, if $f(x) = 3x$ and $g(x) = 2x^2$, $(f+g)(x) = 2x^2 + 3x$. That's it. The only tricky part is managing the domain of both functions.

In the example from just now, the domain of $f(x)$ and $g(x)$ are identical: $x \in \mathbb{R}$ or $-\infty < x < \infty$. But that's not always true. Let's look at another example.

If we redefine the function f to be $f(x) = \sqrt{x}$ instead and try adding the functions again, we get $(f+g)(x) = 2x^2 + \sqrt{x}$. The domain of $(f+g)(x)$ has changed because we've introduced a function that has a domain restriction.

f is a radical function and, with how I've defined f, it has a domain of $x \in [0, \infty)$ (i.e. $0 \leq x < \infty$). Because $g(x)$ still has no restrictions, the domain of $(f+g)(x)$ just happens to be the same as the domain of f in this particular example.

I'll plug values into f and those same values into $(f+g)(x)$ to illustrate this.

x	$f(x) = \sqrt{x}$	$(f+g)(x) = 2x^2 + \sqrt{x}$
2	$\sqrt{2}$	$8 + \sqrt{2}$
1	1	2
0	0	0
-1	$undefined$	$undefined$

$(f+g)(x)$ is undefined at $x = -1$ because it has a radical term, which is undefined for negative terms. The same restriction exists for rational functions. If $f(x) = \dfrac{1}{x}$ instead of $\sqrt{x}$, then the new domain would be $x \neq 0$.

These initial examples have been easy because at least one of the functions has no restrictions. But what if both functions have restrictions? In a case like that, the domains intersect.

If we redefine g to be $g(x) = \dfrac{1}{x}$ and f to be $f(x) = \sqrt{x}$, then $(f+g)(x) = \sqrt{x} + \dfrac{1}{x}$. The domain of f has not changed. It's $x \in [0, \infty)$. The domain of g is $x \neq 0$ because division by 0 is undefined.

The easiest way to visualize the domain of a combination like this is to create a table. Let's create another table to illustrate.

x	$f(x) = \sqrt{x}$	$g(x) = \dfrac{1}{x}$	$(f+g)(x) = \sqrt{x} + \dfrac{1}{x}$
2	$\sqrt{2}$	$\dfrac{1}{2}$	$\sqrt{2} + \frac{1}{2}$
1	1	1	2
0	0	$undefined$	$undefined$
-1	$undefined$	-1	$undefined$

Notice that if *either* of the functions are undefined, $(f+g)(x)$ is also undefined. In this particular example, $(f+g)(x)$ has a domain of $x \in (0, \infty)$.

Definition

Given two functions f and g, their sum $(f+g)(x)$ can be defined as

$$(f+g)(x) = f(x) + g(x).$$

If the function f has a domain of D_1 and the function g has a domain of D_2, the domain of $(f+g)(x)$ is the set of all x values such that $x \in D_1$ and $x \in D_2$ or, more simply, $D_1 \cap D_2$ (intersection of D_1 and D_2).

3.2 Function Subtraction

Function subtraction is identical to addition, except you're subtracting functions instead of adding them.

Definition

Given two functions f and g, their difference $(f-g)(x)$ can be defined as

$$(f-g)(x) = f(x) - g(x).$$

If the function f has a domain of D_1 and the function g has a domain of D_2, the domain of $(f-g)(x)$ is the set of all x values such that $x \in D_1$ and $x \in D_2$ or, more simply, $D_1 \cap D_2$ (intersection of D_1 and D_2).

Do you think you can extend the ideas discussed in Function Addition to Function Subtraction? Give this problem a shot.

Problem

Given two functions f and g defined as $f(x) = \sqrt{x+1}$ and $g(x) = 2x$, find $(f-g)(x)$ and list its domain.

Solution. We're given two functions f and g and asked to subtract g from f. If we apply what we learned when adding functions, we know that the solution to this is

$$(f-g)(x) = \sqrt{x+1} - 2x.$$

To find the domain of this function, let's first find the domain of f and g. The function f is a radical function. That means the radicand (the term inside the square root symbol) must be positive or 0. To find what values of x satisfy this restriction, we can set the radicant equal to 0 and solve for x.

$$\begin{aligned} x + 1 &= 0 \\ x &= -1 \end{aligned}$$

Therefore, the domain of f is $x \in [-1, \infty)$.

The function g, however, has no restriction at all. It's just a polynomial term. That means its domain is $x \in \mathbb{R}$. Because the domain of f is *in* the domain of g, the domain of $(f - g)(x)$ is just the domain of f. Thus, the domain of $(f - g)(x)$ is $x \in [-1, \infty)$.

Nice. Let's do a slightly more difficult one.

Problem

Given that $f(x) = \sqrt{3x^2 + 1}$ and $g(x) = \frac{3}{2x-5}$, find $(g - f)(x)$ and list its domain.

Solution. There are a few things to note with this problem. First, pay attention to the fact that we're subtracting f from g this time $(g - f)$ and not g from f $(f - g)$. Whereas addition is communitative (i.e. the order doesn't matter), subtraction is not. So it's really important that we label this correctly.

$$(g - f)(x) = g(x) - f(x) \implies (g - f)(x) = \frac{3}{2x - 5} - \sqrt{3x^2 + 1}$$

Now that we've made the function, we can determine its domain. I'll work with g first, but it doesn't matter which one you start with.

g is a rational function (fraction), meaning its denominator cannot equal 0. We can find what value of x is invalid by setting the denominator to 0 and solving for x.

$$\begin{aligned} 2x - 5 &= 0 \\ 2x &= 5 \\ x &= \frac{5}{2} \end{aligned}$$

So the domain of g is $x \neq \frac{5}{2}$.

The function f is a radical function, meaning the radicand must be positive or 0. We can determine what values of x satisfy this condition by setting the radicand equal to an inequality that is greater than or equal to 0.

$$\begin{aligned}3x^2 + 1 &\geq 0\\3x^2 &\geq -1\\x^2 &\geq -\frac{1}{3}\end{aligned}$$

We can stop here because we cannot take the square root of both sides of the equation. $\sqrt{-\frac{1}{3}}$ is undefined. This also means that *any* value of x makes this inequality valid. The function f has no domain restriction (i.e. $x \in \mathbb{R}$). When we take a closer look at the function we can see why that is the case.

Because f has no domain restriction, the intersection of the domains of f and g is just the domain of g. Therefore, the domain of $(f - g)(x)$ is $x \neq \frac{5}{2}$.

3.3 Function Multiplication

Function multiplication is a little different from addition and subtraction, but it's the same principle.

Definition

Given two functions f and g, their product $(f \cdot g)(x)$ can be defined as

$$(f \cdot g)(x) = f(x) \cdot g(x).$$

If the function f has a domain of D_1 and the function g has a domain of D_2, the domain of $(f \cdot g)(x)$ is the set of all x values such that $x \in D_1$ and $x \in D_2$ or, more simply, $x \in D_1 \cap D_2$.

Because multiplication and, later, division are a bit different than addition and subtraction, I'll do some example problems before giving one for you to try. Let's consider the functions $f(x) = 3x$ and $g(x) = \frac{12}{13x}$. From the provided definition, we know that

$$(f \cdot g)(x) = f(x) \cdot g(x).$$

Substituting the values for $f(x)$ and $g(x)$ gives

$$(f \cdot g)(x) = 3x \cdot \frac{12}{13x} = \frac{36}{13}$$

Finding the domain of this function is identical to how we did it for addition and subtraction. f is a polynomial, so it has no domain restriction (i.e. $x \in \mathbb{R}$), and g is a rational function, so the denominator cannot equal 0.

$$\begin{aligned} 13x &= 0 \\ x &= 0 \end{aligned}$$

The domain of g is $x \neq 0$. Therefore, the domain of $(f \cdot g)(x)$ is $x \neq 0$.

This is a strange result, though. If $(f \cdot g)(x)$ is a constant (i.e. has no variables), how can there be a domain restriction? There's no x for there to be a restriction.

The problem lies in how we achieved that constant. Notice that, while simplifying $(f \cdot g)(x)$, we canceled out an x. I'll write the step more clearly here:

$$\begin{aligned} 3x \cdot \frac{12}{13x} &= \frac{3}{1} \cdot \frac{x}{1} \cdot \frac{12}{13} \cdot \frac{1}{x} \\ &= \left(\frac{12}{13} \cdot \frac{3}{1}\right)\left(\frac{x}{1} \cdot \frac{1}{x}\right) \\ &= \frac{12(3)}{13} \cdot \frac{x}{x} \\ &= \frac{36}{13} \cdot 1 \\ &= \frac{36}{13} \end{aligned}$$

Although we're left with a constant, the function originally had a variable in the denominator. Canceling that x out doesn't just remove that domain restriction. If either of the functions have a domain restriction, the function composition (multiplication in this case) also has a restriction, regardless of whether or not the variables are canceled.

This is an example of a *point of discontinuity*, which I'll explain in more depth later in the section on continuity.

Try solving a multiplication problem yourself. :)

Problem

Given the functions $f(x) = \sqrt{x}$ and $g(x) = 3x$, find the function composition $(f \cdot g)(x)$ and state its domain.

Solution. From the definition, we know that $(f \cdot g)(x)$ is just $f(x)$ multiplied by $g(x)$. That gives $(f \cdot g)(x) = 3x\sqrt{x}$. To find the domain, we'll need to look at f and g individually.

f is a polynomial function. It, therefore, has no restrictions. g is a radical function, meaning that the radicand cannot equal 0. Creating an inequality with the radicand and zero gives $x \geq 0$.

Since f has no restrictions, but g does, the domain of $(f \cdot g)(x)$ has the same restriction of g. For $(f \cdot g)(x)$, $x \in [0, \infty]$.

Let's try one more problem. This one will be more difficult, but the steps for solving will be the same.

Problem

Given two functions $f(x) = \frac{2x^2}{3x+1}$ and $g(x) = \sqrt{3x^2 - 1}$, find $(f \cdot g)(x)$ and state its domain.

Solution. First, let's multiply the two functions together.

$$(f \cdot g)(x) = \frac{2x^2}{3x+1}\sqrt{3x^2 - 1}$$

Now let's look at the domain of each function. I'll start with f. The function f is a rational function, which means the denominator cannot be equal to 0. So I'll set the denominator equal to 0 and solve for x.

$$\begin{aligned} 3x + 1 &= 0 \\ 3x &= -1 \\ x &= -\frac{1}{3} \end{aligned}$$

So the doamin of f is $x \neq -\frac{1}{3}$.

Now for g. The function g is a radical function, meaning the radicand cannot be less than 0. So I can set the radicand equal to 0 and solve for x.

$$\begin{aligned}
3x^2 - 1 &= 0 \\
3x^2 &= 1 \\
x^2 &= \frac{1}{3} \\
x &= \pm\frac{1}{\sqrt{3}} \\
x &= \pm\frac{\sqrt{3}}{3} && \longleftarrow \text{Rationalized Fraction}
\end{aligned}$$

We're looking for a minimum value for g, so we'll use $x = -\frac{\sqrt{3}}{3}$. This is, indeed, the smallest value of x that satisfies the equation.

$$\begin{aligned}
3\left(-\frac{\sqrt{3}}{3}\right)^2 - 1 &= 0 \\
3\left(\frac{3}{9}\right) - 1 &= 0 \\
\frac{9}{9} - 1 &= 0 \\
1 - 1 &= 0 \\
0 &= 0
\end{aligned}$$

That means the domain of g is $x \in \left[-\frac{\sqrt{3}}{3}, \infty\right)$.

With the domain of both functions found, let's look for the intersection of the domains. f has a domain of $x \neq -\frac{1}{3}$ and g has a domain of $x \in \left[-\frac{\sqrt{3}}{3}, \infty\right)$. The domain of g includes $-\frac{1}{3}$. But, because of that a restriction from f, there is a hole. Thus, the domain of $(f \cdot g)(x)$ is $x \in \left[-\frac{\sqrt{3}}{3}, -\frac{1}{3}\right) \cup \left(-\frac{1}{3}, \infty\right)$ or $x \geq -\frac{\sqrt{3}}{3} \mid x \neq -\frac{1}{3}$.

3.4 Function Division

Division is different from the other operations in that it's domain restriction isn't just the intersection of the domains of f and g. I'll begin by defining function division, and clarify using an example problem.

Definition

Given two functions f and g, their quotient $\left(\frac{f}{g}\right)(x)$ can be defined as

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

If the function f has a domain of D_1 and the function g has a domain of D_2, the domain of $\left(\frac{f}{g}\right)(x)$ is the set of all x values such that $x \in D_1$, $x \in D_2$, **AND** $g(x) \neq 0$.

Definition 1:

$$\text{Domain of } \left(\frac{f}{g}\right)(x) = \{x \mid x \in D_1 \cap D_2 \text{ and } g(x) \neq 0\}$$

Definition 2 (Beginner Friendly):

If the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in [a, b]$, where a and b are finite, real x values (i.e. not infinity and not imaginary) at the edges (boundary) of the domain, then the following is true:

- If $g(x) = 0$ at some value $x = c$, where c is between a and b, then the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in [a, c) \cup (c, b]$ (all values between a and b, except c, are included).
- If $g(x) = 0$ at the boundary a, then the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in (a, b]$ (all values between a and b, excluding a, are included).
- If $g(x) = 0$ at the boundary b, then the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in [a, b)$ (all values between a and b, excluding b, are included).

Let's look at an example problem to see how this definition works. Take two functions $f(x) = 3$ and $g(x) = x$. We want to find $\left(\frac{f}{g}\right)(x)$. So, we divide like so:

$$\left(\frac{f}{g}\right)(x) = \frac{3}{x}.$$

Finding the domain of this, at first, is the same as we normally do it. f is a constant so it has a domain of $x \in \mathbb{R}$. g is a polynomial function, so it, too, has a domain of $x \in \mathbb{R}$. However, notice that the domain of $\left(\frac{f}{g}\right)(x)$ is **not** $x \in \mathbb{R}$. It has an x in the denominator.

It's easy to see here that $x \neq 0$. But in more difficult functions where it may not be as easy to tell, you'd find values of x that make the function in the denominator, g in this case, equal to 0. Let's try it now.

$$\begin{aligned} g(x) &= 0 \\ x &= 0 \end{aligned}$$

From this, we can see that the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in (-\infty, 0) \cup (0, \infty)$ or simply $x \neq 0$. Let's look at a slightly more complicated example.

Consider the functions $f(x) = 3$ and $g(x) = x^2 - 1$. Again, we're trying to find $\left(\frac{f}{g}\right)(x)$. Just like before, let's divide f by g to get our composition:

$$\left(\frac{f}{g}\right)(x) = \frac{3}{x^2 - 1}.$$

The function f, in this example, is still a constant. So it has a domain of $x \in \mathbb{R}$. The function g, this time, is a quadratic. But it still has a domain of $x \in \mathbb{R}$. However, it may not be as obvious this time what the domain of the composition is. Let's set g, the denominator, equal to 0 and solve for x.

$$\begin{aligned} g(x) &= 0 \\ x^2 - 1 &= 0 \\ x^2 &= 1 \\ x &= \pm 1 \end{aligned}$$

This time, we get two different answers. The reason we must consider positive **and** negative 1 is because squaring either gives 1,

$$(1)^2 = 1 \cdot 1 = 1$$
$$(-1)^2 = -1 \cdot -1 = 1,$$

therefore making them valid solutions. Since $x = 1$ and $x = -1$ make $g(x) = 0$, both values of x must be excluded from the domain. Therefore, the domain is $x \in (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$, or simply $x \neq 1$ and $x \neq -1$.

Try doing this one by yourself. I'll provide the solution, but just give it a shot.

Problem

Given two functions $f(x) = \frac{2}{x^2-10} + 5$ and $g(x) = 6\sqrt{3x-2}$, find the function composition $\left(\frac{f}{g}\right)(x)$ and state its domain.

Solution. We're given two functions. First, let's create the function by dividing f by g.

$$\left(\frac{f}{g}\right)(x) = \frac{\frac{2}{x^2-10} + 5}{6\sqrt{3x-2}}$$

This looks like a really scary function. But it's actually good to keep it this way because it preserves $f(x)$ and $g(x)$, there's no mystery as to where they went. After finding the domain and range, we can make this function look less scary. First, the domain of f.

The function f is a rational function, meaning its denominator cannot be equal to 0. So, let's set the denominator equal to 0 and solve.

$$x^2 - 10 = 0$$
$$x^2 = 10$$
$$x = \pm\sqrt{10}$$

Both values of x make the denominator 0. The domain of f is $x \in (-\infty, -\sqrt{10}) \cup (-\sqrt{10}, \sqrt{10}) \cup (\sqrt{10}, \infty)$, or simply $x \neq -\sqrt{10}$ and $x \neq \sqrt{10}$.

The function g is a radical function. So we'll set the radicand equal to 0 and find a minimum value for x.

$$\begin{aligned} 3x - 2 &= 0 \\ 3x &= 2 \\ x &= \frac{2}{3} \end{aligned}$$

This tells us that the domain for g is $x \in \left[\frac{2}{3}, \infty\right)$.

Finally, because g is in the denominator, let's find for what values of x satisfy $g(x) = 0$.

$$\begin{aligned} g(x) &= 0 \\ 6\sqrt{3x-2} &= 0 \\ \sqrt{3x-2} &= 0 \\ 3x - 2 &= 0 \\ 3x &= 2 \\ x &= \frac{2}{3} \end{aligned}$$

Notice that we get the same answer as when we set the radicand equal to 0. This is merely a coincidence. It's best practice set each equal to 0 as separate steps to avoid unecessary mistakes.

Notice that the domain of g discards **all** negative numbers. So our domain restriction of $x \neq -\sqrt{10}$ can be ignored. However, the restriction $x \neq \sqrt{10}$ cannot be ignored. $\frac{2}{3}$ is about 0.6667 and $\sqrt{10}$ is about 3.162. So, putting it all together, the domain of $\left(\frac{f}{g}\right)(x)$ is $x \in \left[\frac{2}{3}, \sqrt{10}\right) \cup \left(\sqrt{10}, \infty\right)$ (all values between $\frac{2}{3}$ and ∞ except $\sqrt{10}$).

That was a beast of a problem. If you managed to get through that unscathed, then well done. But don't worry if you didn't. Again, this was a challenging problem.

Although it's not required for the solution, here's how you simplify that function, if you're curious. I'll write it as a separate problem.

Problem

Simplify the expression:

$$\left(\frac{f}{g}\right)(x) = \frac{\frac{2}{x^2-10} + 5}{6\sqrt{3x-2}}$$

Solution. First, let's convert the second term in the numerator such that it has the same denominator as the first term.

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{\frac{2}{x^2-10} + \frac{5(x^2-10)}{x^2-10}}{6\sqrt{3x-2}} \\ &= \frac{\frac{2+5(x^2-10)}{x^2-10}}{6\sqrt{3x-2}} \\ &= \frac{\frac{2+5x^2-50}{x^2-10}}{6\sqrt{3x-2}} \\ &= \frac{\frac{5x^2-48}{x^2-10}}{6\sqrt{3x-2}}\end{aligned}$$

Next, recall that if we have a number a divided by a number b that is divided again by a number c, we can rewrite it as the number a divided by the product of b and c.

$$\left(\frac{f}{g}\right)(x) = \frac{5x^2-48}{(x^2-10)(6\sqrt{3x-2})}$$

Now we move the factor of 6 to the beginning of the denominator in preparation to rationalize it.

$$\left(\frac{f}{g}\right)(x) = \frac{5x^2-48}{6(x^2-10)\sqrt{3x-2}}$$

To rationalize the fraction, we simply multiply the numerator and denominator by the radical factor.

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{5x^2-48}{6(x^2-10)\sqrt{3x-2}} \cdot \frac{\sqrt{3x-2}}{\sqrt{3x-2}} \\ &= \frac{(5x^2-48)(\sqrt{3x-2})}{6(x^2-10)(3x-2)}\end{aligned}$$

This is the final, most simplified form. It still looks a bit scary, but it's a lot less scary than what we had to start with.

Chapter 4

Composite Functions

Function compositions and composite functions sound similar, but they're quite different. Instead of multiplying or adding functions together, we're putting functions inside of other functions. You'll often see the analogy of putting a machine in another machine. Although this may be a helpful visualization to some, I will not be using it. Instead, I'll explain what it is using an example.

4.1 Notation For Function Notation

What I want to do before that, however, is clarify the notation for composite functions. This is often a huge point of confusion for many students, as was the case for me. To avoid that confusion, I'll show both ways of notating composite functions here.

- Notation 1: $f(g(x))$
- Notation 2: $(f \circ g)(x)$

Both are ways of writing composite functions. I find Notation 1 more intuitive, so I will be using it for the remainder of this text. However, if you see $(f \circ g)(x)$, understand that it is exactly the same thing.

4.2 What Is A Composite Function

A composite function is basically a function inside of another function. As mentioned, an example is best suited to explain this. Let's start with something simple.

Given a funtion $f(x) = 3x$, we know that we can plug a value for x and get a cooresponding y value. For instance, if we plug $x = 5$ into f, we'll get $f(5) = 15$. I'll write the process of solving $f(5)$ in more detail:

$$\begin{aligned} f(5) &= 3(5) && \text{Substituting 5 for } x \\ f(5) &= 15 && \text{Multiplying } 5 \times 3 \end{aligned}$$

This may seem trivial, but remembering this will help us move onto composite functions.

Now, along with the function f, I'll define a new function $g(x) = 10x$. What do you think would happen if, instead of plugging a constant for the variable x in the function f, we plugged an entire function? What I mean is this: instead of plugging $x = 5$, what if we plugged $x = g(x)$? Well... Let's try it.

$$\begin{aligned} f(g(x)) &= 3(g(x)) && \text{Substituting } g(x) \text{ for } x \\ f(10x) &= 3(10x) && \text{Substituting the value of } g(x) \\ f(10x) &= 30x && \text{Multiplying } 3 \times 10x \end{aligned}$$

We didn't get a value. Instead, we got a new function entirely. When we composed f and g we got

$$f(g(x)) = 30x.$$

Notice that, when I was solving for $f(g(x))$, I substituted $g(x)$ for $10x$ and wrote $f(10x) = 30x$. Although you *can* do this, it's not recommended unless you're literally asked to plug $10x$ into the function. If you want to find a composite function, it's best to stick with $f(g(x))$ without replacing $g(x)$ for whatever it may be ($10x$ in this case).

The reason you don't want to replace $g(x)$ is because you may be asked to substitute a value into a composite function. For example, you may be asked to find $f(g(12))$.

If you're asked to find something like $f(g(12))$, you treat it just like any other function, You replace x with whatever was plugged in.

$$\begin{aligned} f(g(12)) &= 30(12) && \text{Substituting 12 for } x \\ f(g(12)) &= 360 && \text{Multiplying } 30 \times 12 \end{aligned}$$

I want to show the solution for $f(g(12))$ in a slightly different way. Notice that, in $f(g(12))$, the value $g(12)$ is inside of the function f. So, instead of plugging x directly into $f(g(x))$, I'll plug it into $g(x)$, and then plug *that* value into f.

$$\begin{aligned} g(12) &= 10(12) \\ g(12) &= 120 \end{aligned}$$

Now we can take 120 and plug it into f.

$$\begin{aligned} f(g(12)) &= 3(g(12)) \\ f(g(12)) &= 3(120) \\ f(g(12)) &= 360 \end{aligned}$$

Notice that we get the same answer.

4.3 Domain Restrictions

Like with function compositions, composite function have domains to look out for. A method that I've found to work quite well is the following.

1. Create the function $f(g(x))$ and **DO NOT SIMPLIFY**.
2. Then, find the restrictions like you would a normal function.
3. simplify the function.

Here's an example.

Given the function $f(x) = \sqrt{x}$ and $g(x) = x-2$, find $f(g(x))$ and list its domain.

Following the strategy above, we create the function $f(g(x)) = \sqrt{x-2}$. Now, without doing anything else, we look for domain restrictions.

The only one this composite function has is the radical. Set the radicant equal to 0 and solve for x.

$$\begin{aligned} x - 2 &= 0 \\ x &= 2 \end{aligned}$$

Because this is a radical function, we can say that the domain restriction for $f(g(x))$ is $x \geq 2$ or $x \in [2, \infty)$.

Let's look at a slightly more difficult example this time.

Given the function $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$, find $f(g(x))$ and list its domain.

First, I'll create $f(g(x))$:

$$f(g(x)) = \frac{1}{\frac{1}{x-1}}.$$

Now, I'll look for a domain restriction. We see that there is a denominator. Let's set that equal to zero.

$$\frac{1}{x-1} = 0$$

This is still tricky, so let's go one level deeper and set the denominator of the fraction above equal to 0.

$$x - 1 = 0$$

Now, after solving for x, we can see that the domain restrition for this rational function is $x \neq 1$. From this, we can say that the domain of $f(g(x))$ is $x \neq 1$. Now, we can simplify this.

$$\begin{aligned} f(g(x)) &= \frac{1}{\frac{1}{x-1}} \\ &= 1 \cdot \frac{x-1}{1} \\ &= x-1 \end{aligned}$$

Alternatively, you could use this procedure:

1. Find the domain of $g(x)$. This domain is A.
2. Find the domain of f, but using g instead of x. This is domain B.
3. Take the intersection of the domains A and B.
4. Create $f(g(x))$ and simplify.

This alternateive method may look a bit tricky, so let's do an example together using the previous problem.

Given the function $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$, find $f(g(x))$ and list its domain.

First, I'll take the domain of g. g is a rational function, meaning its denominator cannot equal 0. So set its denominator equal to 0 and solve for x.

$$x - 1 = 0 \Longleftarrow x = 1$$

This means x cannot equal 1 ($x \neq 1$) or $A = (-\infty, 1) \cup (1, \infty)$.

Now we'll find the domain of f. f, too, is a rational function. So let's set its numerator equat to 0 and solve for x.

$$x = 0$$

After finding the domain of f, which is $x \neq 0$ replace the x with the function g.

$$\frac{1}{x-1} \neq 0$$

This statement happens to be true for all values of x because the numerator is non-zero. $\frac{1}{x-1}$ cannot equal 0 anyway. So $B = (-\infty, \infty)$.

Finally, we'll take the intersection of A and B ($A \cap B$).

$$A \cap B = ((-\infty, 1) \cup (1, \infty)) \cap (-\infty, \infty)$$

This looks really scary, but it's easy to visualize with a numberline or just by stating what the restrictions are.

- A says that x cannot be 1.
- B says that x could by any number since $\frac{1}{x-1}$ can never be 0.

So the intersection of these is that x canont be 0. So the domain of $f(g(x))$ is $x \in (-\infty, 1) \cup (1, \infty)$ or just $x \neq 1$.

Simplifying the nested fraction gives the same answer as before.

$$f(g(x)) = x - 1 \text{ where } x \in (-\infty, 1) \cup (1, \infty)$$

Let's do one more example. Given $f(x) = \sqrt{x+3}$ and $g(x) = \frac{1}{x-2}$, find $f(g(x))$ and state its domain.

First, I'll take the domain of g. g is a rational function, so it's denominator cannot equal 0. So I'll set it's denominator equal to 0 and solve for x

$$x - 2 = 0 \Longrightarrow x = 2$$

This means that $x \neq 2$ or $A = (-\infty, 2) \cup (2, \infty)$

Next, I'll find the domain of f. f is a radical function, meaning the radicand must be greater than or equal to 0. So I'll set the radicand equal to 0 to find the minimum value.

$$x + 3 = 0 \Longrightarrow x = -3$$

This means that x must be greater than or equal to -3 ($x \geq -3$) or $B = [-3, \infty)$.

Now we take the intersection of A and B. To make this easier, let's state what each of the domains tell us.

- A tells us that x cannot equal 2 ($x \neq 2$).
- B tells us that x must be greater than or equal to -3 ($x \geq -3$).

So the final domain is $x \in [-3, 2) \cup (2, \infty)$. If interval notation looks a bit intimidating, you can also write it like this: $-3 \leq x < 2$ and $2 < x < \infty$.

Let's formalize composite functions with a definition.

Definition

Given two functions f and g, the composite function $f \circ g$ is the result of substituting the function g into the function f. For functions $f(x)$ and $g(x)$:

$$f \circ g = f(g(x))$$

We can write the domain of $f \circ g$ like so:

$$Domain(f \circ g) = \{x \in Domain(g) \mid g(x) \in Domain(f)\}$$

This means that x must be in the domain of g, and the entire function $g(x)$ must be in the domain of the function f.

This definition is where the second procedure comes from.

Try this problem for size:

Problem

Given the functions $f(x) = \sqrt{3x+1}$ and $g(x) = \frac{1}{x^2-3}$, find the composite function $g(f(x))$ and state its domain.

Solution. I'm gonna use the first procedure for this solution. I'll leave the procedure here as a reference.

1. Create the function $f(g(x))$ and **DO NOT SIMPLIFY**.
2. Then, find the restrictions like you would a normal function.

3. simplify the function.

First, I'll combine the functions. But remember that the problem asked for $g(f(x))$, not $f(g(x))$.

$$g(f(x)) = \frac{1}{\left(\sqrt{3x+1}\right)^2 - 3}$$

Second, I'll find the domain restriction. The outer most function is a rational function (fraction), meaning the denominator cannot be 0. To find the restriction of the rational, I'll set the denominator equal to 0 and solve for x.

$$\begin{aligned}\left(\sqrt{3x+1}\right)^2 - 3 &= 0\\ \left(\sqrt{3x+1}\right)^2 &= 3\\ 3x+1 &= 3\\ 3x &= 2\\ x &= \frac{2}{3}\end{aligned}$$

So we know that $x \neq \frac{2}{3}$.

The denominator of the fraction also has a radical term. The radical term must be greater than or equal to 0. So I'll set the radicand equal to 0 and solve for x to find the minimum value.

$$\begin{aligned}3x+1 &= 0\\ 3x &= -1\\ x &= -\frac{1}{3}\end{aligned}$$

Now we additinally know that $x \geq -\frac{1}{3}$. Combining the two restrictions gives $x \in \left[-\frac{1}{3}, \frac{2}{3}\right) \cup \left(\frac{2}{3}, \infty\right)$ (i.e. all values from $-\frac{1}{3}$ to infinity **except** $\frac{2}{3}$).

Now we can simplify this as best we can.

$$\begin{aligned}g(f(x)) &= \frac{1}{\left(\sqrt{3x+1}\right)^2 - 3}\\ &= \frac{1}{3x+1-3}\\ &= \frac{1}{3x-2}\end{aligned}$$

So our final answer is

$$g(f(x)) = \frac{1}{3x-2} \text{ where } x \in \left[-\frac{1}{3}, \frac{2}{3}\right) \cup \left(\frac{2}{3}, \infty\right)$$

Chapter 5

Inverse Functions

Inverse functions are a direct extention of composite functions because the definition is derived from it. Let's learn how to find an inverse of a function.

To find the inverse of a function, simply rewrite the function as an equation of y, swap the x and y, and solve for y. That's about it. Let's try it with an example.

Problem

Given the function $f(x) = 3x^2$, find it's inverse.

Solution. We're given the function $f(x) = 3x^2$. First, we'll rewrite it as an equation of y.

$$y = 3x^2$$

Then, we'll swap the x and y.

$$x = 3y^2$$

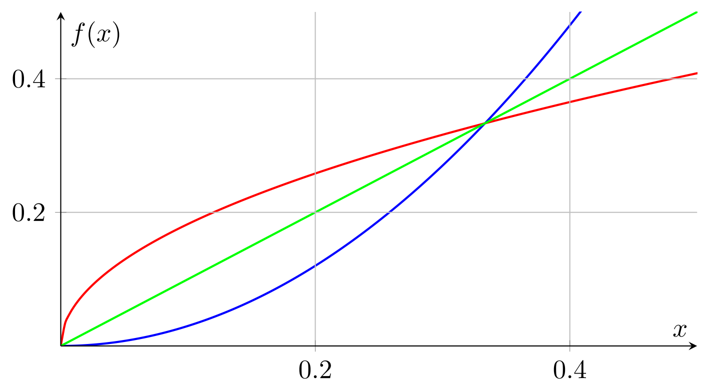
Finally, we'll solve for y.

$$
\begin{aligned}
x &= 3y^2 \\
\frac{x}{3} &= y^2 \\
\sqrt{\frac{x}{3}} &= y
\end{aligned}
$$

We can now rewrite this as the inverse function of f.

$$f^{-1}(x) = \sqrt{\frac{x}{3}}$$

Notice how the graphs of f (blue) and f^{-1} (red) intersect in the first quadrant of the xy-plane.



They are symmetrical along the line $y = x$ (green). Creating a table for each of these functions side by side reveals yet another property that inverse functions have.

Let's substitute $x = 3$ into f. That gives:

$$
\begin{aligned}
f(3) &= 3(3)^2 \\
&= 3(9) \\
&= 27
\end{aligned}
$$

Now, let's plug this value into the inverse function f^{-1}.

$$\begin{aligned} f^{-1}(3) &= \sqrt{\frac{27}{3}} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Let's try the same for a different value, like 5.

$$\begin{aligned} f(5) &= 3(5)^2 \\ &= 3(25) \\ &= 75 \end{aligned} \qquad\qquad \begin{aligned} f^{-1}(75) &= \sqrt{\frac{75}{3}} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

What this means is: plugging the y-value of f into f^{-1} gives the x value of f. A cleaner way of demonstrating this is by creating tables of each function side by side.

x	$f(x) = 3x^2$
1	3
2	12
3	27

x	$f^{-1}(x) = \sqrt{\frac{x}{3}}$
3	1
12	2
27	3

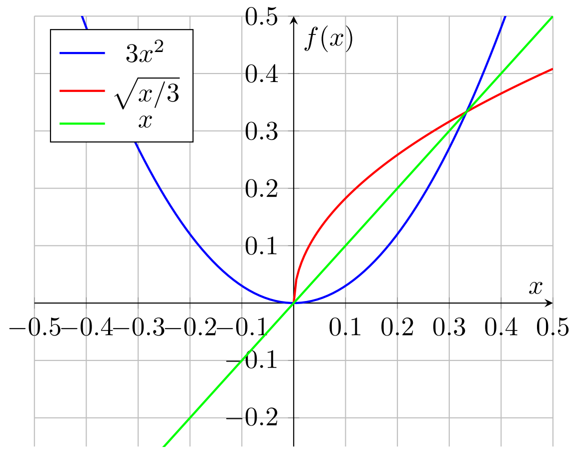
You may have noticed at this point that I haven't included the left side of the parabola. That's because the left side is **not** actually part of the inverse. The inverse function $f^{-1} = \sqrt{\frac{x}{3}}$ is a radical, meaning the result of f^{-1} are valid. If I use negative x values for f, I literally cannot swap the x and y. The x in $f(x)$ is being square, which negates any negative.

x	$f(x) = 3x^2$
-1	3
-2	12
-3	27

x	$f^{-1}(x) = \sqrt{\frac{x}{3}}$
3	1 (not negative!!)
12	2 (not negative!!)
27	3 (not negative!!)

If I include the left side of the parabola in the graph I presented before, you can see why this is true.

Figure 5.1: Parabola-Radical Example



The function f^{-1} (red line) does not intersect f (blue line) **or** $y = x$ (green line) when $x < 0$.

Let's formalize this with a definition.

Definition

Given a function f, it's inverse f^{-1} is such that

$$f(f^{-1}) = x \text{ and } f^{-1}(f(x)) = x$$

and the domain of f is equal to the range of f^{-1} (and vise versa).

Before giving a problem for you to try on your own, I'll solve one more, slightly more difficult problem.

Given the function $f(x) = 3x^2+2x-1$, find its inverse f^{-1} and state its domain.

First, I'll rewrite it as an equation of y and swap the variables

$$\begin{aligned} y &= 3x^2 + 2x - 1 \\ x &= 3y^2 + 2y - 1 \end{aligned}$$

Now I'll solve for y.

$$\begin{aligned} 0 &= 3y^2 + 2x - 1 - x \\ 0 &= 3y^2 + 2x - (1 + x) \end{aligned}$$

I'll use the quadratic formula on the above equation with a, b, and c defined like so

$$\begin{aligned} a &= 3 \\ b &= 2 \\ c &= -(1 + x) = -x - 1 \end{aligned}$$

The reason I'm using the quadratic formula is because we're given the standard form of a quadratic: ax^2+bx+c. In the same way you'd solve for x by using the quadratic formula, we can solve for y using the quadratic formula. The solution for y is as follows.

$$\begin{aligned}
y &= \frac{-(2) \pm \sqrt{(2)^2 - 4(3)(-x-1)}}{2(3)} && \text{Applied Quadratic Formula} \\
&= \frac{-2 \pm \sqrt{4 + 12(x+1)}}{6} && \text{Simplified Radicand and Denominator} \\
&= \frac{-2 \pm \sqrt{4 + 12x + 12}}{6} && \text{Distributed 12 Into The Parentheses} \\
&= \frac{-2 \pm \sqrt{12x + 16}}{6} && \text{Added Like Terms} \\
&= \frac{-2 \pm \sqrt{4(3x + 4)}}{6} && \text{Factored Out 4 (the perfect square)} \\
&= \frac{-2 \pm 2\sqrt{3x + 4}}{6} && \text{Took } \sqrt{4} = 2 \\
&= \frac{-1 \pm \sqrt{3x + 4}}{3} && \text{Divided Top and Bottom By 2}
\end{aligned}$$

$$y = \frac{-1 \pm \sqrt{3x+4}}{3}$$

In this case, we're only interested in the positive solution for y for a few reasons:

1. We want to ensure that the resulting expression is actually a function. Leaving it as $\pm$ turns the parabola sideways, which fails the vertical line test.

2. The negative solution introduces a complication that I'll explain in just a moment.

That leaves us with this inverse function as our solution.

$$\boxed{f^{-1}(x) = \frac{\sqrt{3x+4}-1}{3}} \qquad \text{Swapped the terms in the numerator (optional)}$$

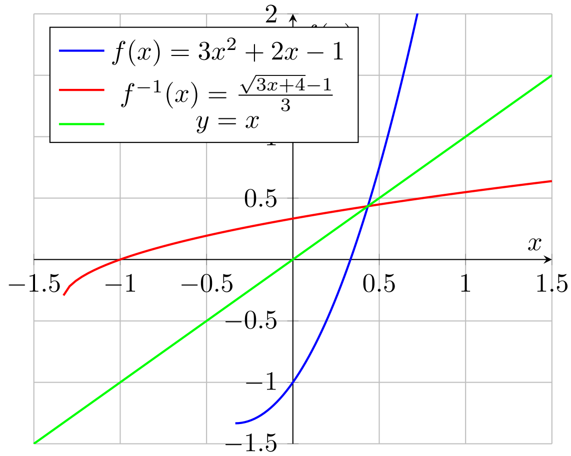
Now for the domain. Although this *is* a rational function, there are no variables in the denominator, so there is no restriction there. So the only limitation is the radical. We need to find the minimum value (0) of the radicand, so we can set it to 0 and solve for x.

$$\begin{aligned} 3x + 4 &= 0 \\ 3x &= -4 \\ x &= -\frac{4}{3} \end{aligned}$$

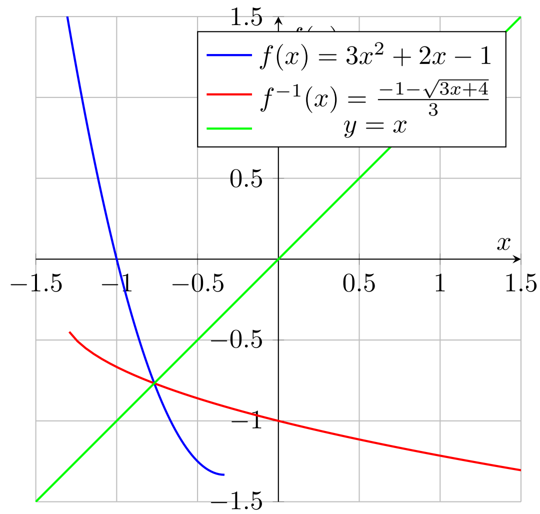
This means $x \geq -\frac{4}{3}$. Since this is the only restriction, we're done. To make it a bit more formal, we can write the domain restriction in interval notation like so: $x \in \left[-\frac{4}{3}, \infty\right)$.

Now. About that complication I was talking about... In our solution, we only cared about the positive solution. But you actually *can* use the negative. The reason you're able to do that goes back to the example with the radical and parabola (see Figure 5.1).

To illustrate this, I'll create a graph for $f(x) = 3x^2 + 2x - 1$ and $f^{-1}(x) = \frac{\sqrt{3x+4}-1}{3}$. Notice how these are symmetric about the line $y = x$.



Now look at the graph for $f(x) = 3x^2 + 2x - 1$ and $f^{-1}(x) = \frac{-1-\sqrt{3x+4}}{3}$.



Notice how these are also symmetric about the line $y = x$. The only difference is that the intersection is in the third quadrant instead of the first.

Although both are equally valid, the reason we take the positive solution is purely because of convention. Mathematicians came across this problem a while ago and had to decide which they wanted to normalize. They chose the positive solution as the convention, so that's what we use now.

I urge you to stick with the convention. Using the negative solution might not fare super well with your teacher (i.e. you may lose points for not choosing the positive).

As promised, I'll now give a probelm for you to try.

Problem

Given the function $g(x) = \frac{3}{2x+1}$, find the inverse function g^{-1} and state its domain.

Solution. First, let's rewrite this function as an equation of y in terms of x.

$$y = \frac{3}{2x+1}$$

Next, let's swap the variables.

$$x = \frac{3}{2y+1}$$

Now, let's solve for y.

$$\begin{aligned} x(2y+1) &= 3 && \text{Multiplied both sides by } 2y+1 \\ 2y+1 &= \frac{3}{x} && \text{Divided both sides by } x \\ 2y &= \frac{3}{x} - 1 && \text{Subtrated both sides by } 1 \\ y &= \frac{\frac{3}{x} - 1}{2} && \text{Divided both sides by } 2 \end{aligned}$$

Since the denominator at the very bottom (2) doesn't have any variables, I can distribute it into the numerator. Doing so gives your inverse function.

$$g^{-1}(x) = \frac{3}{2x} - \frac{1}{2}$$

Now to find the inverse. The only limitation here is the denominator $2x$. Since that denominator cannot equal 0, we can set it equal to 0 to see what value of x would make the fraction invalid.

$$
\begin{aligned}
2x &= 0 \\
x &= 0 && \text{Divided both sides by 2}
\end{aligned}
$$

Although, we could've just noticed that anything times 0 is 0, meaning $2x = 0$ would simplify to $x = 0$, without doing any of that work. Since this is as simplified as it gets, we have our final answer:

$$g^{-1}(x) = \frac{3}{2x} - \frac{1}{2} \text{ where } x \neq 0$$

Chapter 6

Piecewise Functions

Piecewise functions are a kind of break from the composite and inverse functions we've been working on up to this point. Piecewise functions are best used when the formula deciding the input changes after a certain x value. Examples often used with piecewise functions include tax brackets, shipment/delivery price changes, discounts dependent on age, etc.

Before hopping into any examples, I think it's best to get acquainted with what a piecewise function looks like and how to graph it. Consider the below piecewise function:

$$f(x) = \begin{cases} x & 0 \leq x \leq 4 \\ -(x-4)+4 & 4 < x \leq 8 \\ \frac{x-8}{2} & x > 8 \end{cases}$$

What do you think this looks like? I've written the explanation of this on the next page, so take a moment to think about what happens before moving on.

- Why are there compound inequalities next to equations?
- Why are some of the inequalities $<$ and others $\leq$?
- What's up with that $x > 8$ in the bottom row? Why is it different?
- Could these compound inequalities by written in some other way?

6.1 Describing Piecewise Functions

Alright. Here it is.

The $f(x)$ is the name of the function, as is the case with any other function. It is set equal to a curly bracket containing cases.

Each row first identifies and equation and next by an interval that the equation applies within.

Using the above function as an example:

$$x \qquad 0 \leq x \leq 4$$

means to say that the equation $y = x$ applies for all x values between 0 and 4.

$$-(x-4)+4 \qquad 4 < x \leq 8$$

means to say that the equation $y = -(x-4)+4$ applies for all x values between 4 and 8 excluding 4. The value $x = 4$ is excluded because it's already accounted for in the previous domain.

$$\frac{x-8}{2} \qquad x > 8$$

means to say that the equation $\frac{x-8}{2}$ applies for all x values that are greater than 8. Again, the inequality is not $x \geq 8$ because the value $x = 8$ was accounted for in the previous domain.

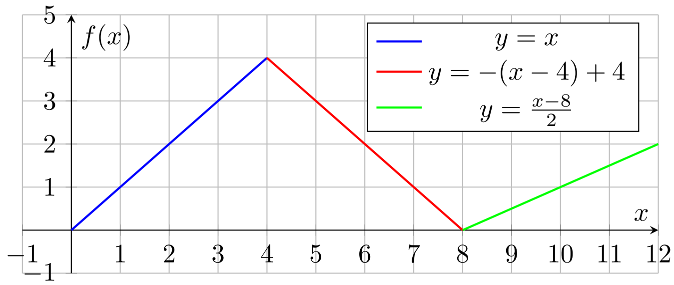
The inequality $x > 8$ could be written as $8 < x < \infty$ so that it matches the other inequalities.

The piecewise function could also be rewritten using interval notation, like so:

$$f(x) = \begin{cases} x & x \in [0,4] \\ -(x-4)+4 & x \in (4,8] \\ \frac{x-8}{2} & x \in (8,\infty) \end{cases}$$

This piecewise function produces the graph below:

Figure 6.1: Graph Of Piecewise Function $f(x)$



6.2 Example with Piecewise Functions

Now that we know what a piecewise function *is*, let's look at an example of where it may be used.

"Totally real bank name" gives money out for free. If we say that x represents the number of hours that have passed, then:

- The first 4 hours, represented as $0 \leq x \leq 4$, \$10 are given out per hour.
- The following 9 hours, represented as $4 < x \leq 13$, \$7 are given per hour.
- Every hour after that, represented as $x > 13$ or $13 < x < \infty$, \$5 are given out per hour.

Instead of writing each and every case, we can use a piecewise function to set all three cases equal to a function f. This function will output the amount of money given out after x hours.

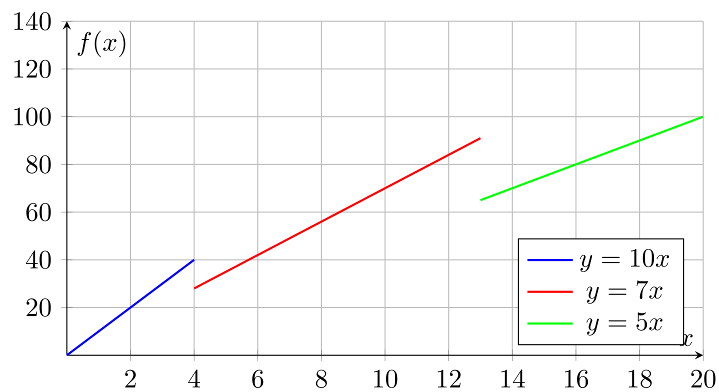
6.2.1 Potential Trap

We should be careful when constructing such a function, however. We can't simply call the equatoins $10x$, $7x$ and $5x$.

$$f(x) = \begin{cases} 10x & x \in [0, 4] \\ 7x & x \in (4, 13] \\ 5x & x \in (13, \infty) \end{cases} \qquad \longleftarrow \text{Bad Piecewise Function}$$

Doing so creates the piecewise function above and graph in Figure 6.2 below.

Figure 6.2: Piecewise Function with Incorrect Equations



Just throwing these into a piecewise function would create a valid function, but one that doesn't represent the situation given. For it to represent the provided situation, the lines need to be connected. Let's examine why.

If 4 hours passed, and 10 dollars were given out every hour, then 40 dollars were given out.

When the fifth hour passes, the piecewise function moves to the next equation, which we've currently assigned as $7x$. Plugging $x = 5$ to $7x$ gives $7 \times 5 = 35$. That's 5 dollars less than what was given out in the first 4 hours.

Since the amount given out in the fifth hour must add to the amount already given on the first 4, we expect the amount after 5 hours (i.e. $f(5)$) to be 47 dollars.

You may think, "just shift the function up 12 units ($y = 7x + 12$). Doing that gets us up to the 47 we expect ($35 + 12 = 47$)." Although this *would* work, it requires that you know how much was given out at the fourth hour.

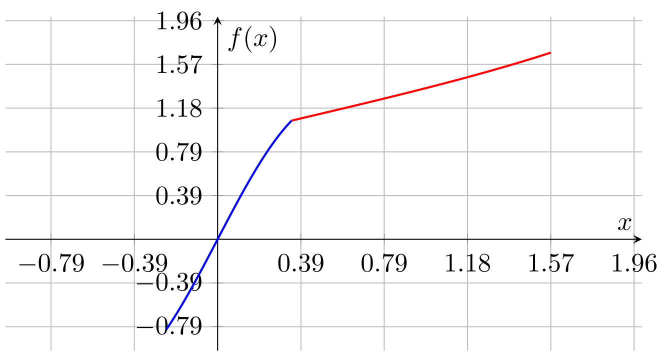
Of course you can do that in this example. It's easy. But something like the one below would not be very easy to determine:

- $\sin(\pi x) + \tan\left(\frac{x}{2}\right)$ acts on $x \in \left[-\frac{\pi}{13}, \frac{\pi}{9}\right]$
- $\tan\left(\frac{\pi^2 x}{22}\right)$ acts on $x \in \left(\frac{\pi}{9}, \frac{\pi}{2}\right]$

The solution to that looks like this if you're curious:

$$g(x) = \begin{cases} \sin(\pi x) + \tan\left(\frac{x}{2}\right) & x \in \left[-\frac{\pi}{13}, \frac{\pi}{9}\right] \\ \tan\left(\frac{\pi^2\left(x - \frac{\pi}{9}\right)}{22}\right) + \sin\left(\frac{\pi^2}{9}\right) + \tan\left(\frac{\pi}{18}\right) & x \in \left(\frac{\pi}{9}, \frac{\pi}{2}\right] \end{cases}$$

Figure 6.3: Super Hard Problem (Scale: $\frac{\pi}{8}$)



6.2.2 The Correct Way To Do It

Let's go back to the problem we were working on.

Instead of predicting or calculating how much we should add or subtract, we can literally move the entire function using vertical and horizontal transformations.

In this case, we know that $y = 10x$ up until $x = 4$.

We're already given that $x = 4$, and $10 \times 4 = 40$, meaning $y = 40$.

What's left is to apply those transformations (4 right and 40 up) to $7x$.

That gives: $y = 7(x - 4) + 40$.

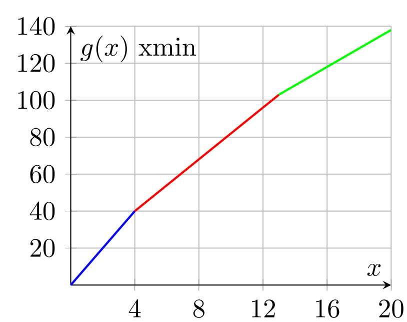
This method does not rely on any calculations. All you need is the right-most edge of the bound (4 in this case) and the value of the equation at that bound (40 in this case). The same can be done for the last function.

In the first 4 hours, the bank gave out 40 dollars. In the next 9 hours, the bank gave out $7 \times 9 = 63$ dollars. That means the total amount given is $40 + 63 = 103$ dollars.

We also know that the right-most boundary of the second equation is $x = 13$. That means we need to transform $5x$ 13 units to the right and 103 units up, giving $y = 5(x - 13) + 103$.

Putting that all together gives the piecewise function and graph below:

$$g(x) = \begin{cases} 10x & x \in [0, 4] \\ 7(x-4) + 40 & x \in (4, 13] \\ 5(x-13) + 103 & x \in (13, \infty) \end{cases}$$



6.3 Jump Discontinuity

Up to this point, we've been working with functions that continue cleanly from one another. To illustrate this point, I'll use another example. For those of you also taking physics, this sort of example may look familiar:

Problem

Janet is a frequent runner and takes the same route to her friend's house every day. On her daily runs, she:

- Runs $1000\,m$ in 30 minutes.
- Rests for 5 minutes.
- Runs $750\,m$ in 30 minutes.
- Rests for 10 minutes.
- Runs $1200\,m$ in 45 minutes.

Create a piecewise function and a Velocity vs. Time graph (VT Graph) representing Janet's velocity.

For those who haven't taken physics, a Velocity vs. Time graph is just like a normal function but the x-axis is time t and the y-axis is velocity v.

Also, instead of using $f(x)$ or $g(x)$, I'll be using $v(t)$, a velocity function with time (in minutes) as the independent variable.

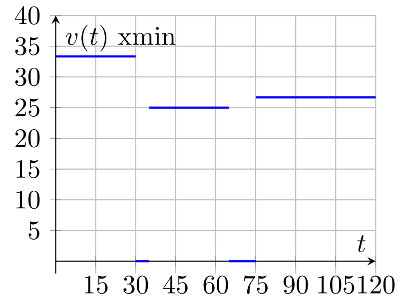
We're not given any velocities, but we can find them quickly. Velocity is calculated by dividing the distance traveled by the time it took to travel that distance. For example, if you drove 30 miles in 2 hours then you were going $30\,mi/2\,hr = 15\,mi/hr$.

Using the equation $v = \frac{\Delta s}{\Delta t}$ (s stands for change in position btw), we get:

$$\begin{array}{lllll} v_1 = \dfrac{1000}{30} & v_2 = \dfrac{0}{5} & v_3 = \dfrac{750}{30} & v_4 = \dfrac{0}{10} & v_5 = \dfrac{1200}{45} \\ v_1 = \dfrac{100}{3} & v_2 = 0 & v_3 = 25 & v_4 = 0 & v_5 = \dfrac{80}{3} \end{array}$$

We don't have any variables here, so there's nothing to line up. That means we can create the piecewise function and graph right now:

$$v(t) = \begin{cases} \frac{100}{3} & t \in [0, 30] \\ 0 & t \in (30, 35] \\ 25 & t \in (35, 65] \\ 0 & t \in (65, 75] \\ \frac{80}{3} & t \in (75, 120] \end{cases}$$



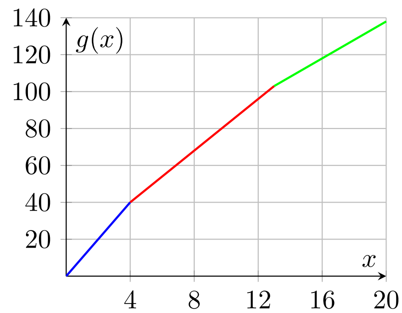
This function doesn't smoothly transition like with the bank example. It breaks off and continues in different places. It jumps from $v = 35$ all the way down to $v = 0$, hence why it's called a "jump" discontinuity. This is **not** continuous. Also... 25 minutes per minute? Dayum she's slow. But that's beside the point.

6.4 Removable Discontinuity

A removable discontinuity is a less obvious than a jump discontinuity. The function may *look* continuous, but it'll have a hole somewhere. I won't spend too much time on this because it's solved almost identically to regular piecewise functions. In fact, I can actually use a previous example to demonstrate this.

Here was the result from that bank example:

$$g(x) = \begin{cases} 10x & x \in [0,4] \\ 7(x-4)+40 & x \in (4,13] \\ 5(x-13)+103 & x \in (13,\infty) \end{cases}$$



This function is continuous because there are no gaps. But what if we changed the inequality $x \in [0,4]$ to $x \in [0,4)$?

If we did that, the function would begin at 0, continue all the way up to $3.99999\ldots$, then skip 4 to go to $4.00000\ldots1$ and continue as normal. Making some substitutions would help visualize what's going on.

$$\begin{aligned} f(3.9) &= 39 \\ f(4) &= undefined \\ f(4.1) &= 40.7 \end{aligned}$$

Since the function has nothing defined for $x = 4$ (i.e. it's skipped), it's labeled as undefined. The function f has no value at $x = 4$. On a graph, that is usually labeled by an open circle.

With that change, the piecewise function and graph now looks like this:

$$g(x) = \begin{cases} 10x & x \in [0, 4) \\ 7(x-4)+40 & x \in (4, 13] \\ 5(x-13)+103 & x \in (13, \infty) \end{cases}$$